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(54) **METHOD AND APPARATUS FOR  
PERFORMING CONGESTION CONTROL IN  
NR V2X**

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**H04B 7/0417** (2017.01)

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(52) **U.S. Cl.**

CPC ..... **H04W 74/0808** (2013.01); **H04B 7/0417**  
(2013.01); **H04B 7/061** (2013.01);

(Continued)

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CPC ... H04W 72/25; H04W 72/40; H04W 36/253;  
H04B 7/061; H04B 7/0417; H04L  
27/2646

See application file for complete search history.

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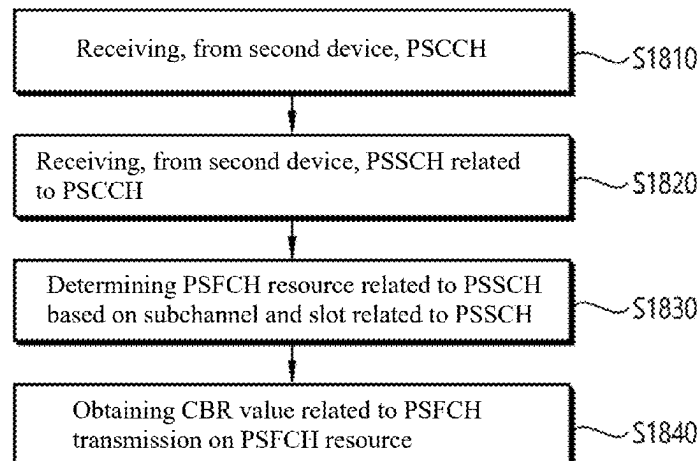
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(74) *Attorney, Agent, or Firm* — LEE, HONG,  
DEGERMAN, KANG & WAIMEY

(57) **ABSTRACT**

A method for performing wireless communication by a first  
apparatus and an apparatus for supporting same are pro-  
vided. The method may comprise the steps of: receiving a  
physical sidelink control channel (PSCCH) from a second  
apparatus; receiving a physical sidelink shared channel  
(PSSCH) related to the PSCCH from the second apparatus;  
determining a physical sidelink feedback channel (PSFCH)  
resource related to the PSSCH, on the basis of a slot and  
sub-channel related to the PSSCH; and obtaining a channel  
busy ratio (CBR) value related to PSFCH transmission on  
the PSFCH resource. The CBR value may be obtained on the  
basis of (i) a first slot before a first offset from a slot that  
received the PSSCH related to the PSFCH resource, and (ii)

(Continued)



measurement in a time section between the first slot and a second slot before a second offset.

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### 17 Claims, 24 Drawing Sheets

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**H04B 7/06** (2006.01)  
**H04W 24/10** (2009.01)  
**H04W 72/0446** (2023.01)  
**H04W 72/20** (2023.01)  
**H04W 72/25** (2023.01)  
**H04W 74/0808** (2024.01)  
**H04L 27/26** (2006.01)  
**H04W 4/40** (2018.01)  
**H04W 52/28** (2009.01)

#### (52) U.S. Cl.

CPC ..... **H04W 24/10** (2013.01); **H04W 72/0446**  
(2013.01); **H04W 72/20** (2023.01); **H04W**  
**72/25** (2023.01); **H04W 72/40** (2023.01);  
**H04L 27/2646** (2013.01); **H04W 4/40**  
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FIG. 1  
--PRIOR ART--

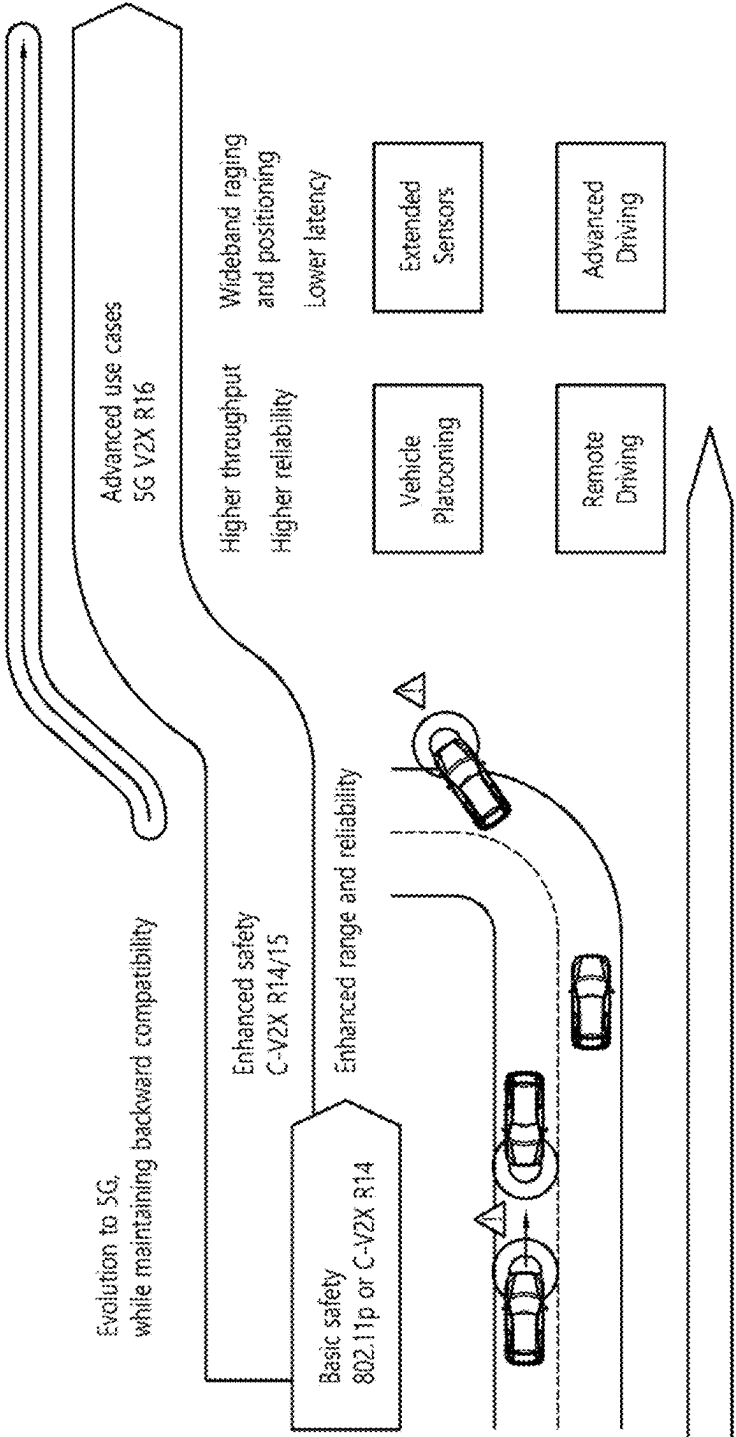


FIG. 2

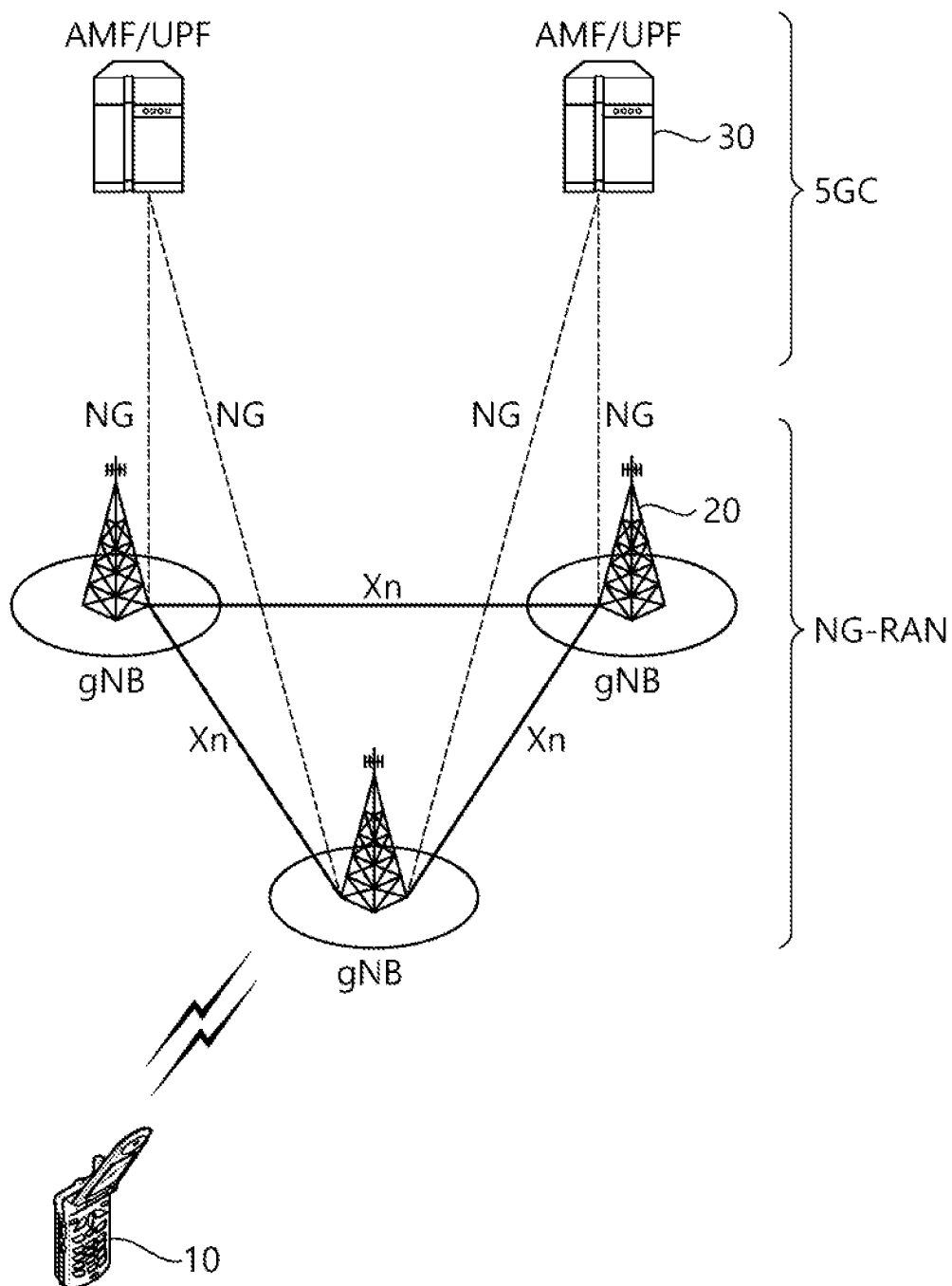


FIG. 3

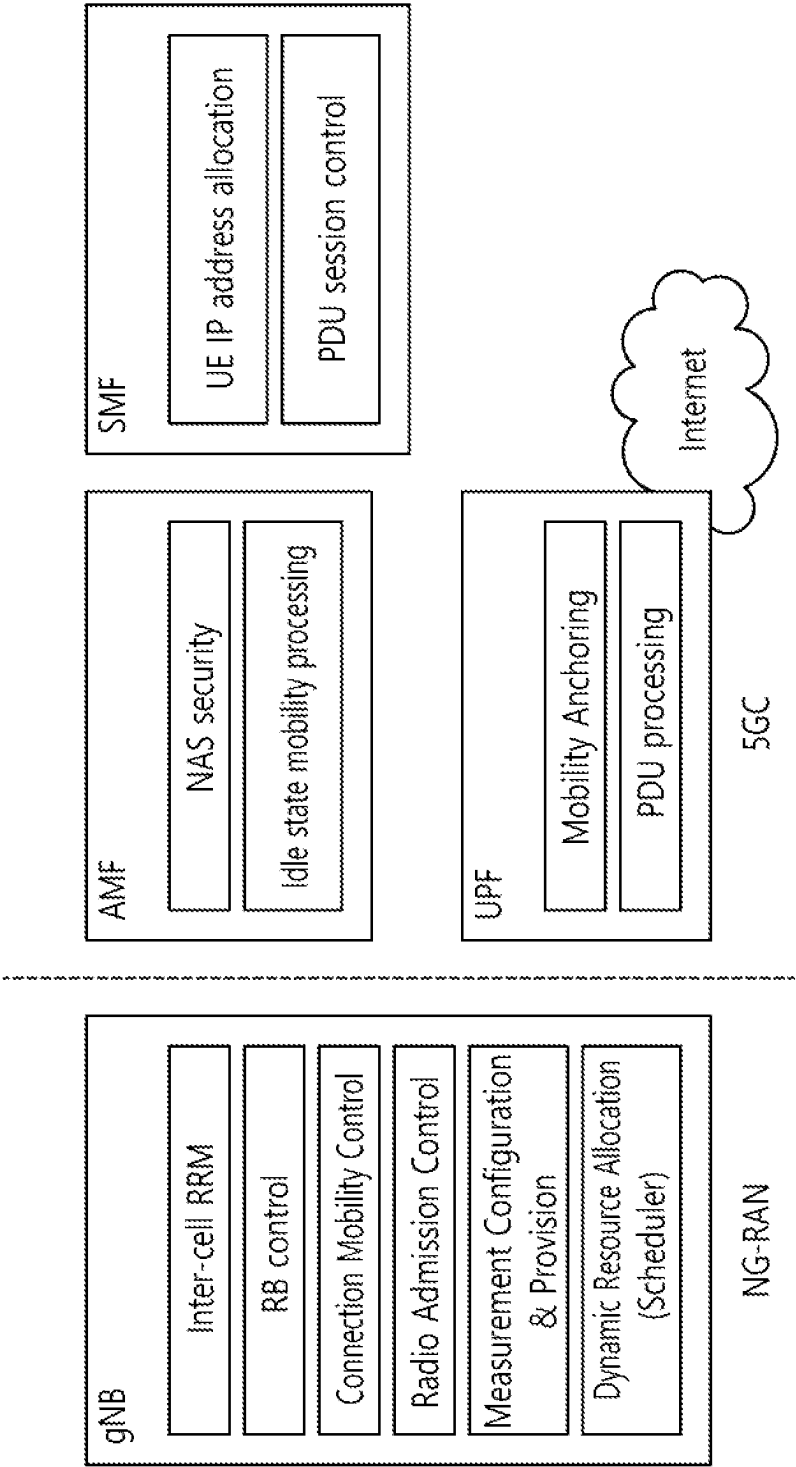


FIG. 4

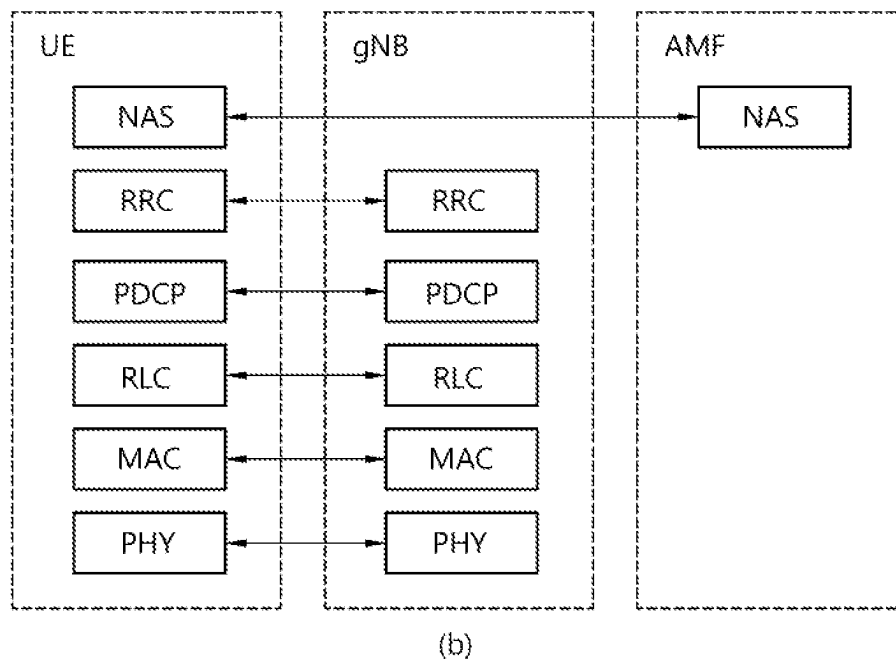
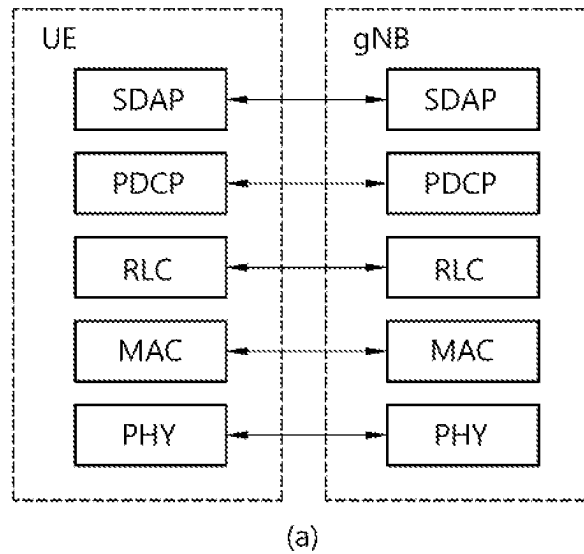


FIG. 5

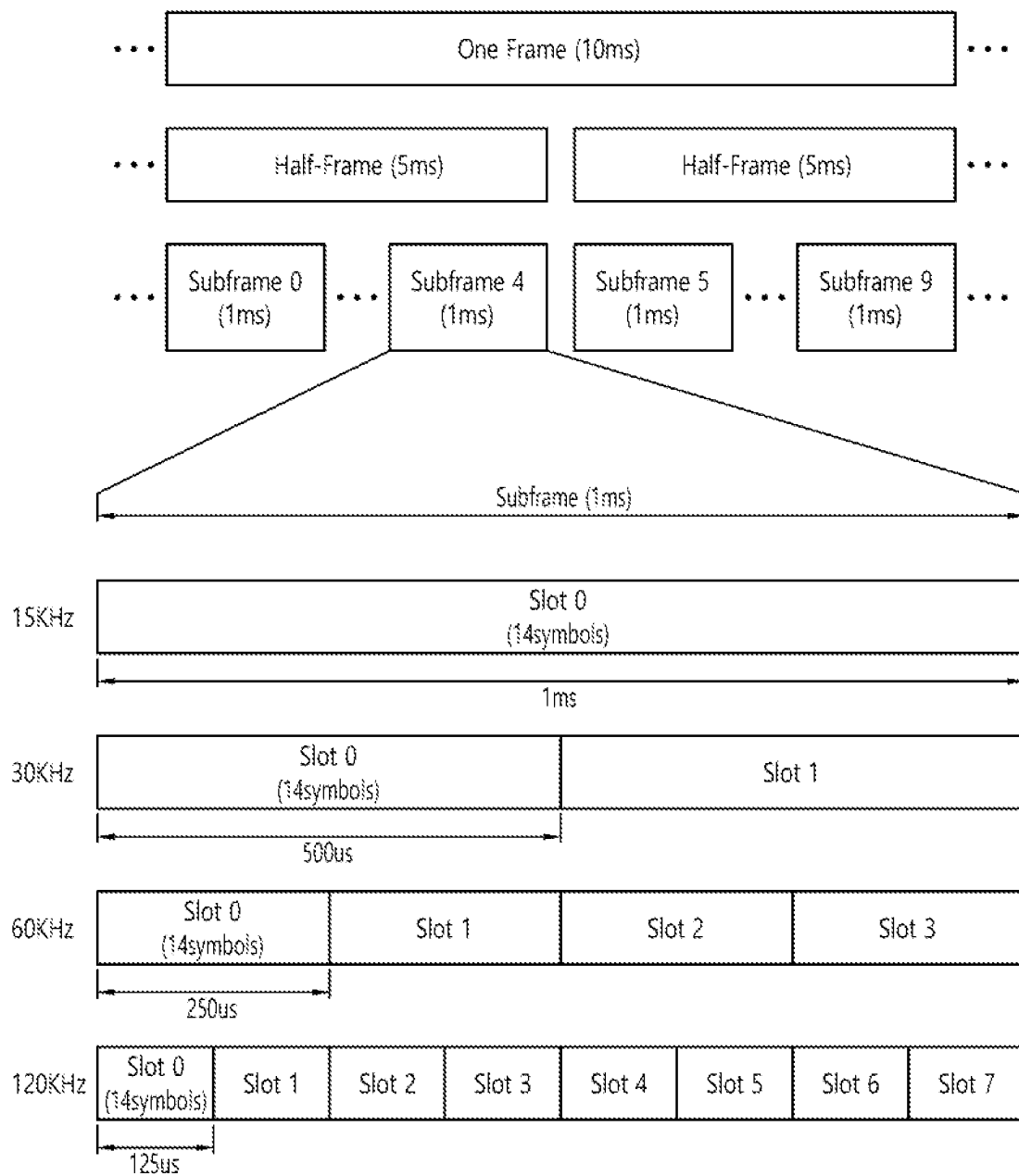


FIG. 6

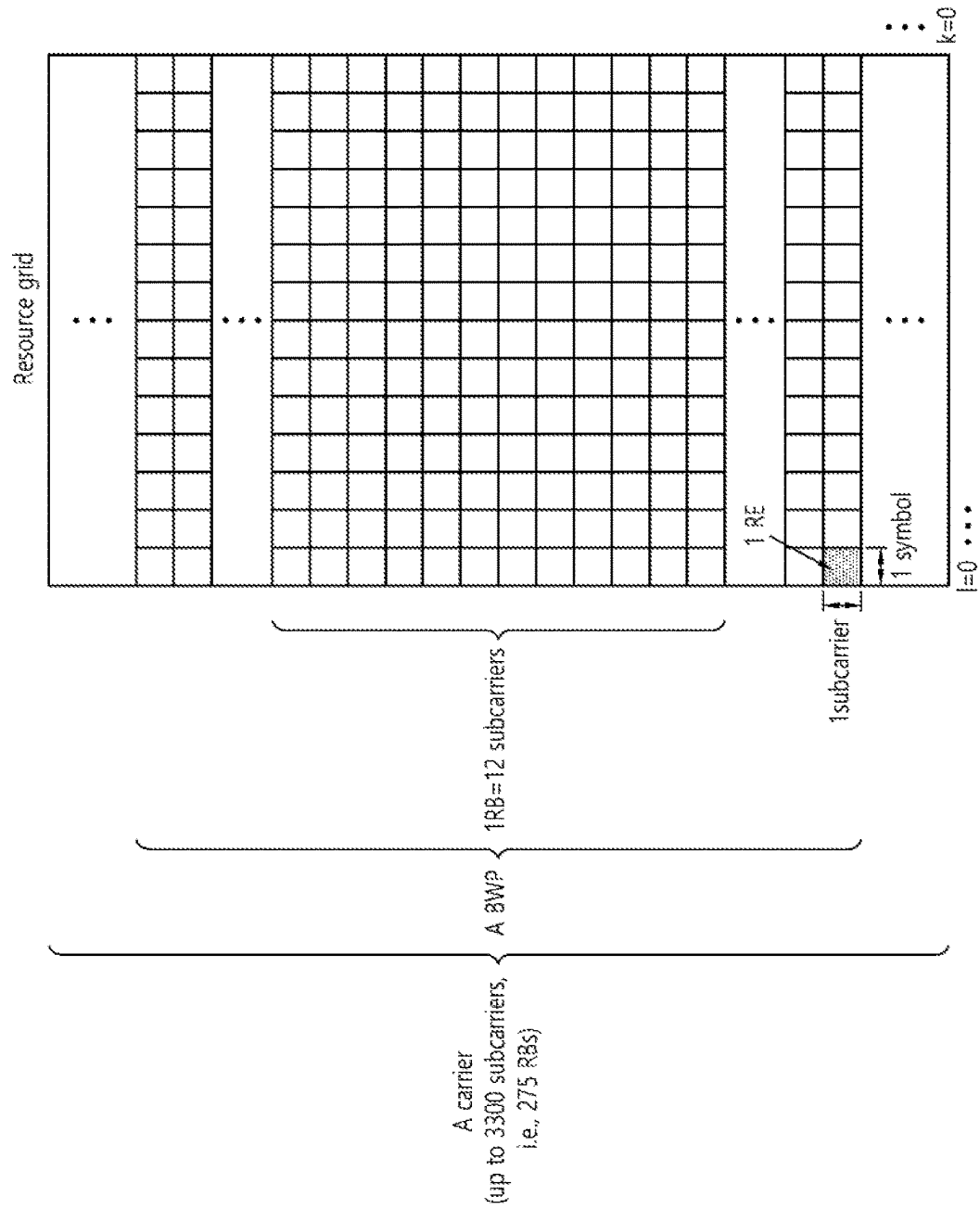




FIG. 7

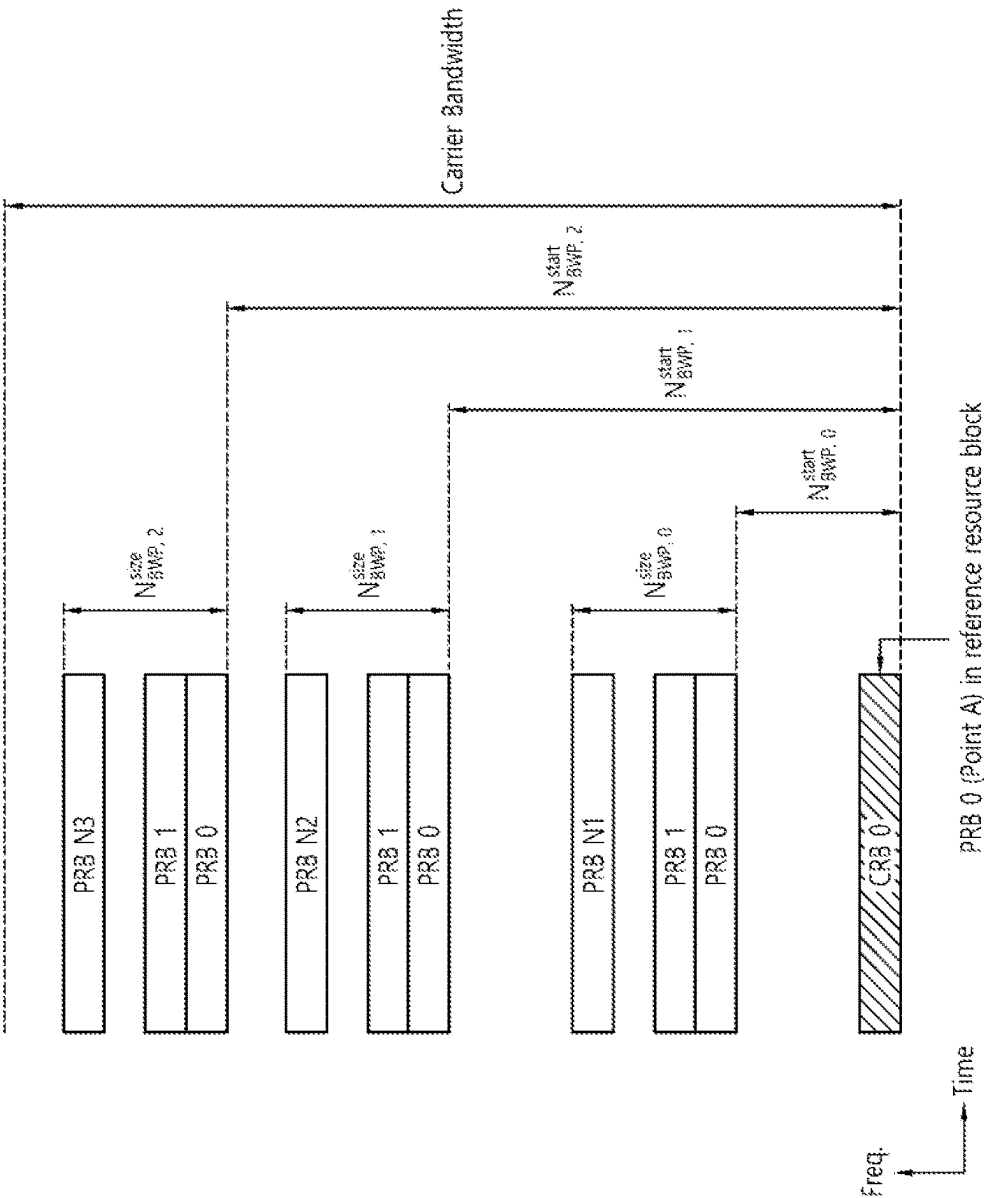
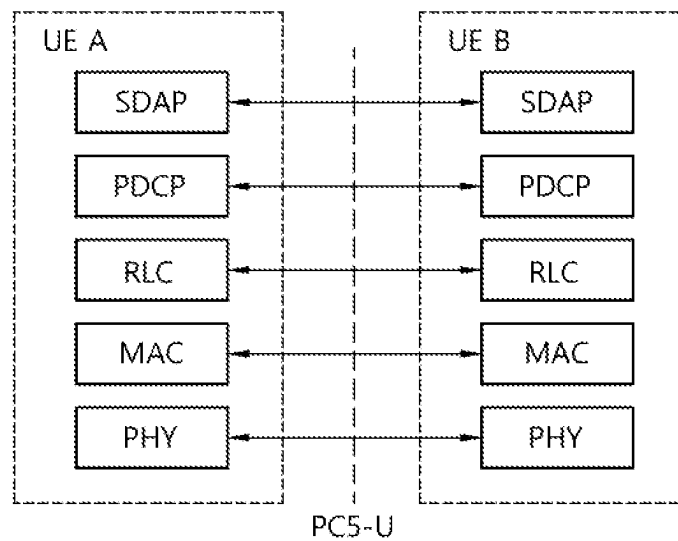
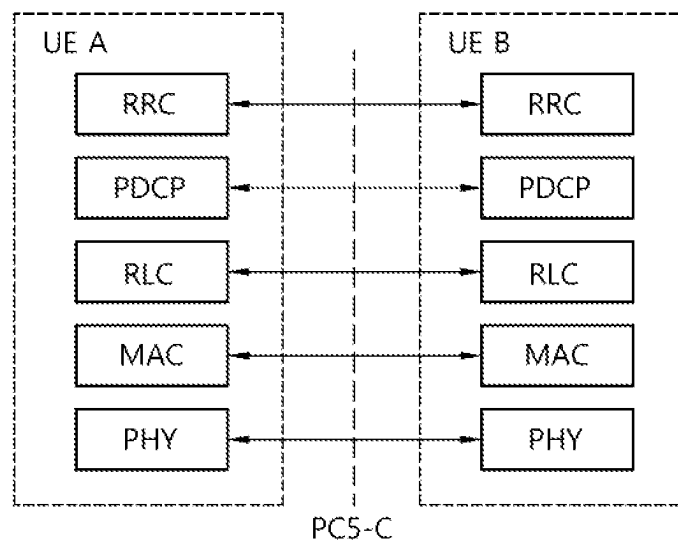


FIG. 8



(a)



(b)

FIG. 9

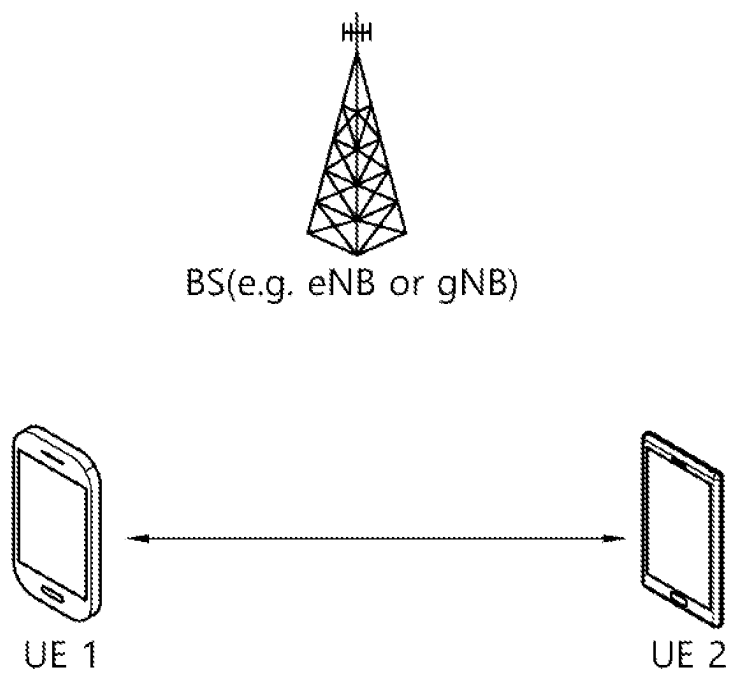


FIG. 10

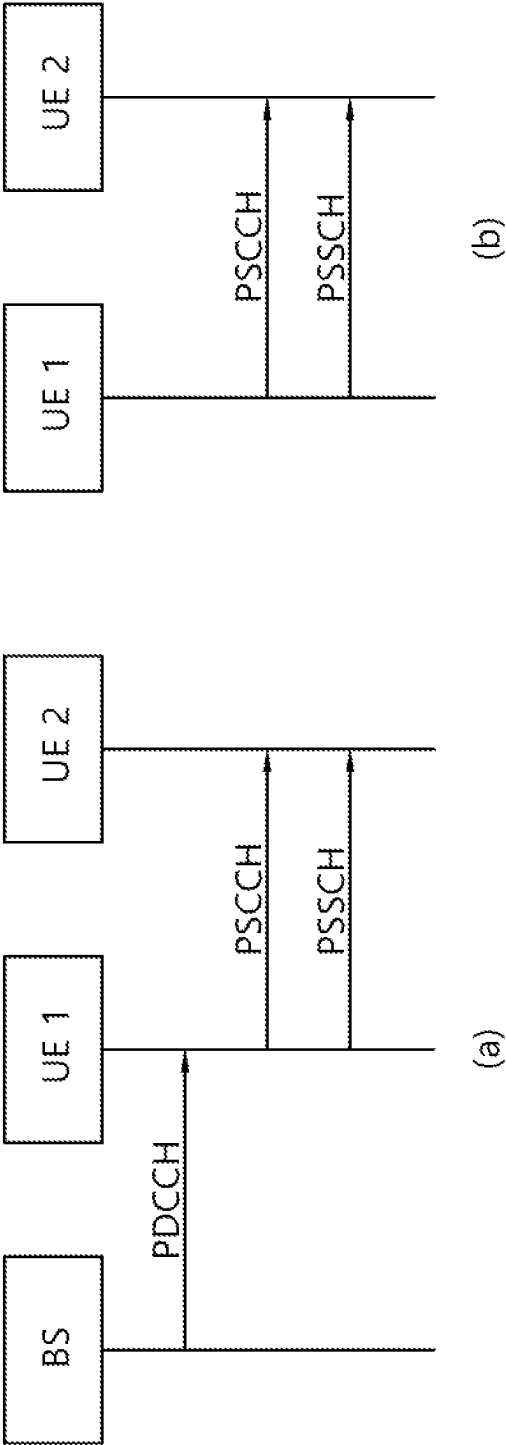


FIG. 11

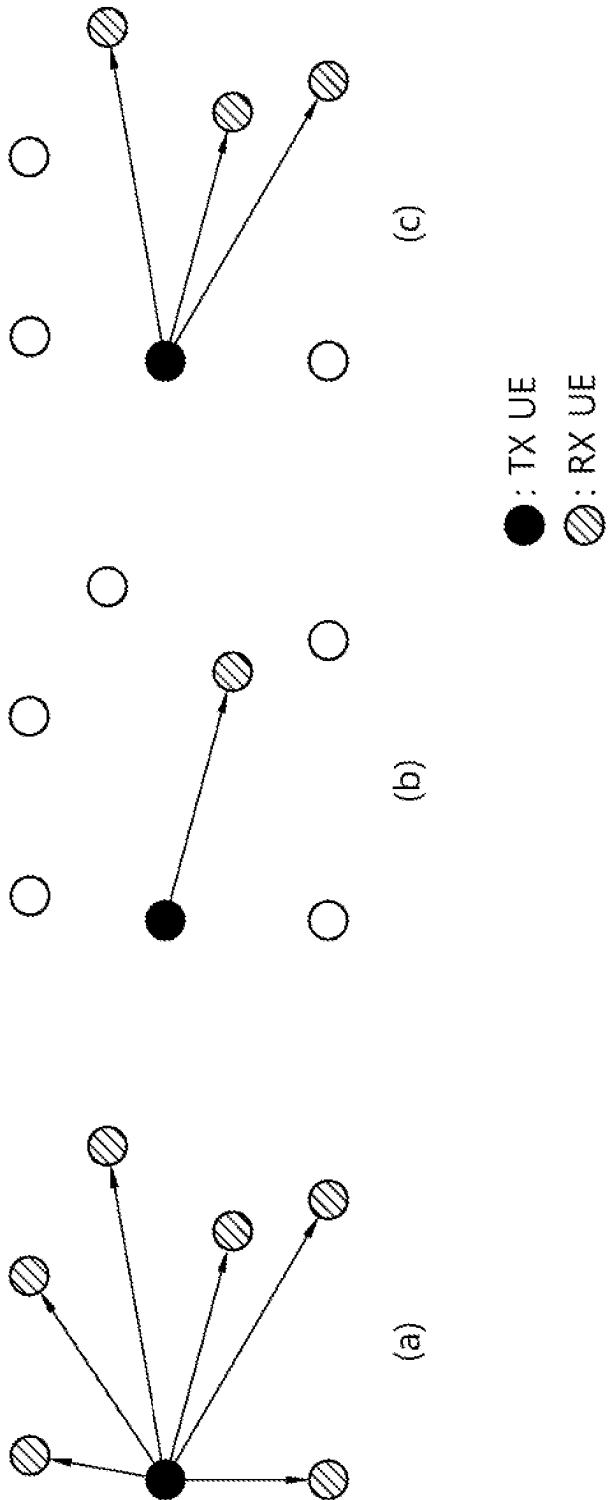


FIG. 12

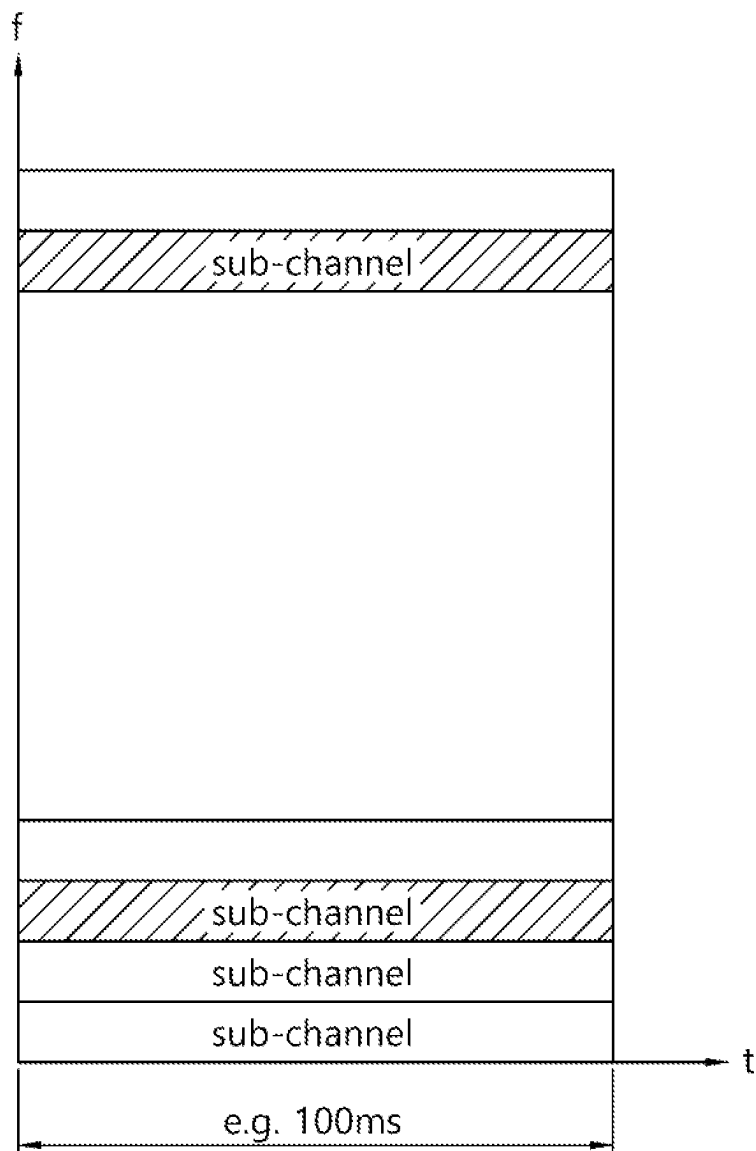


FIG. 13

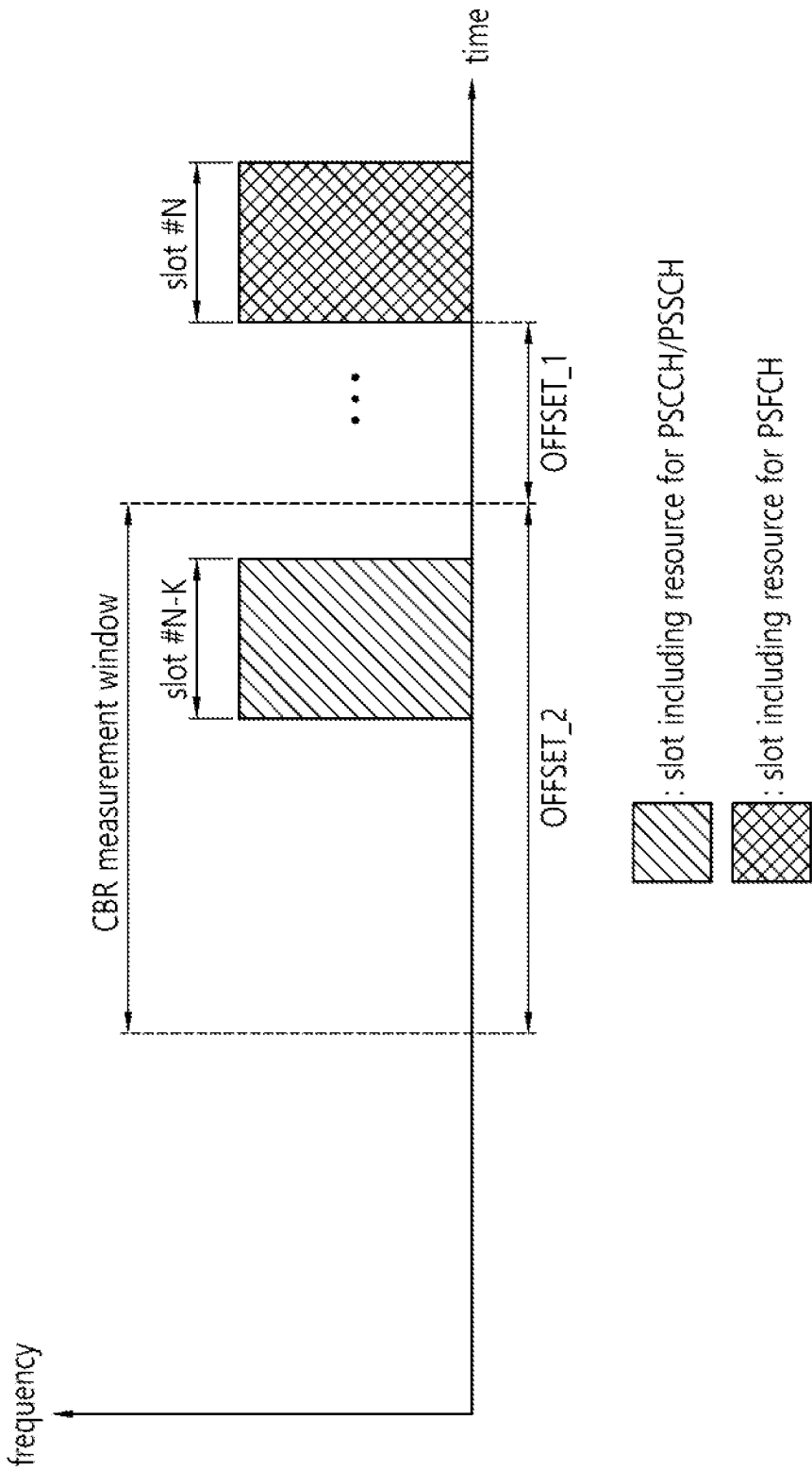


FIG. 14

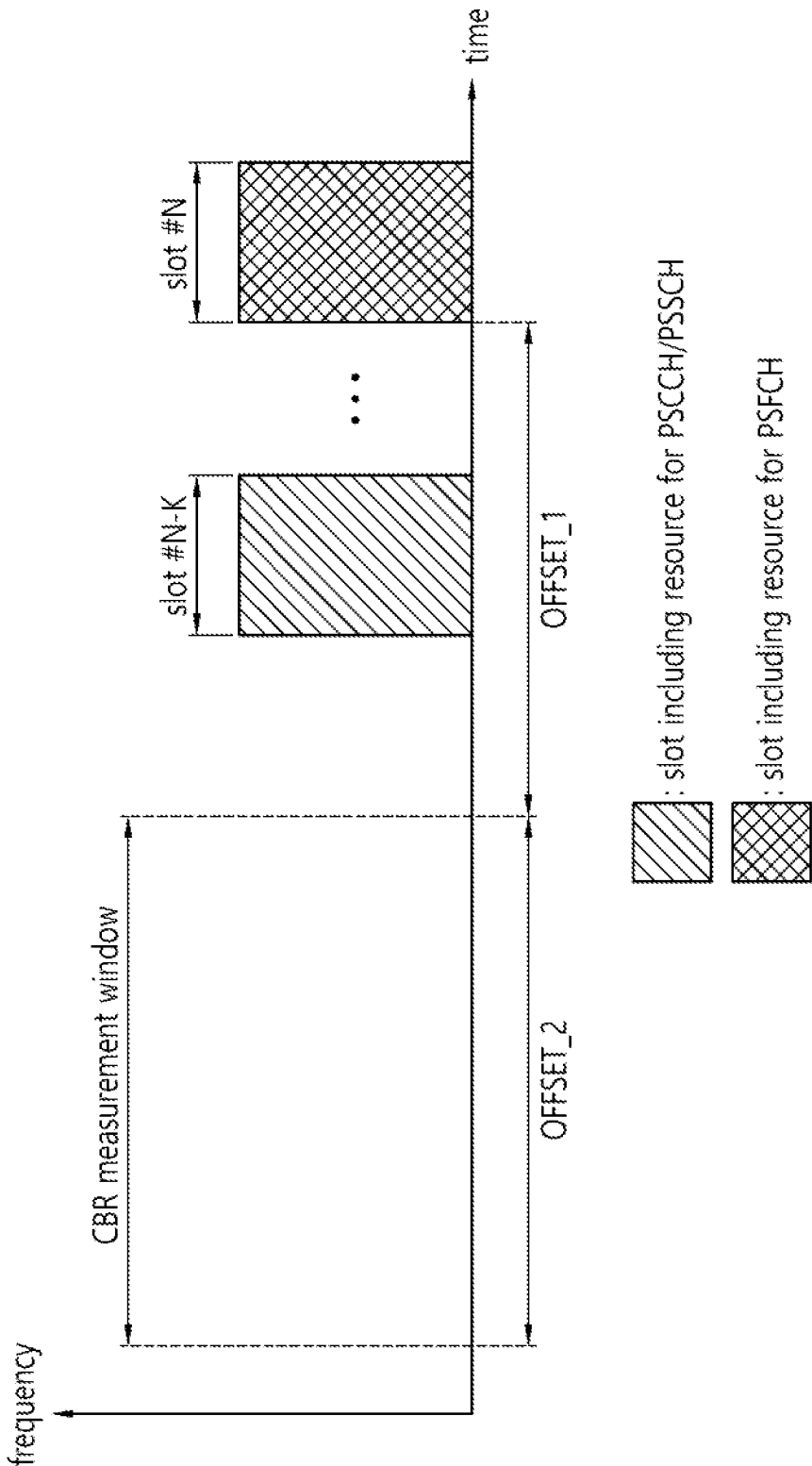




FIG. 15

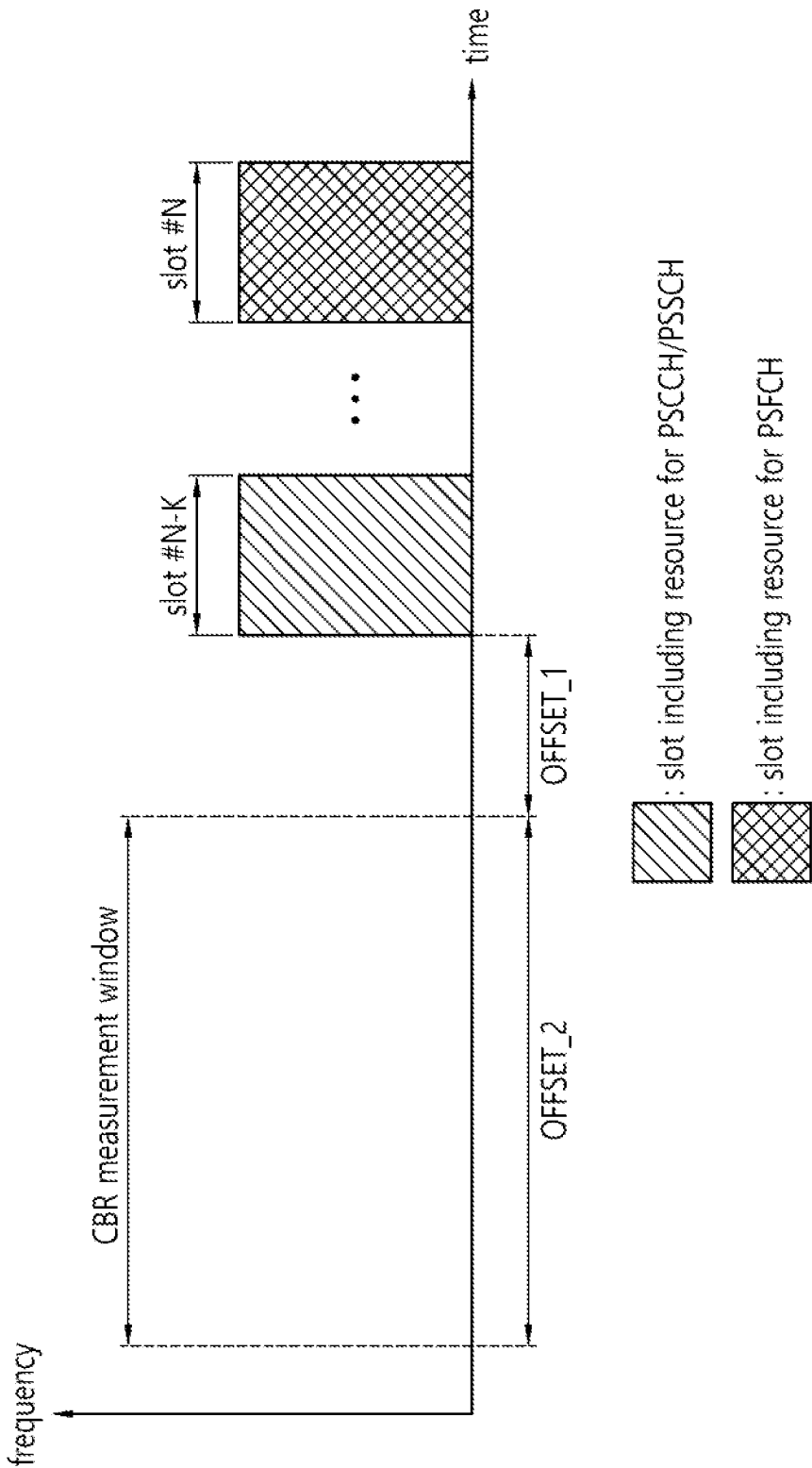


FIG. 16

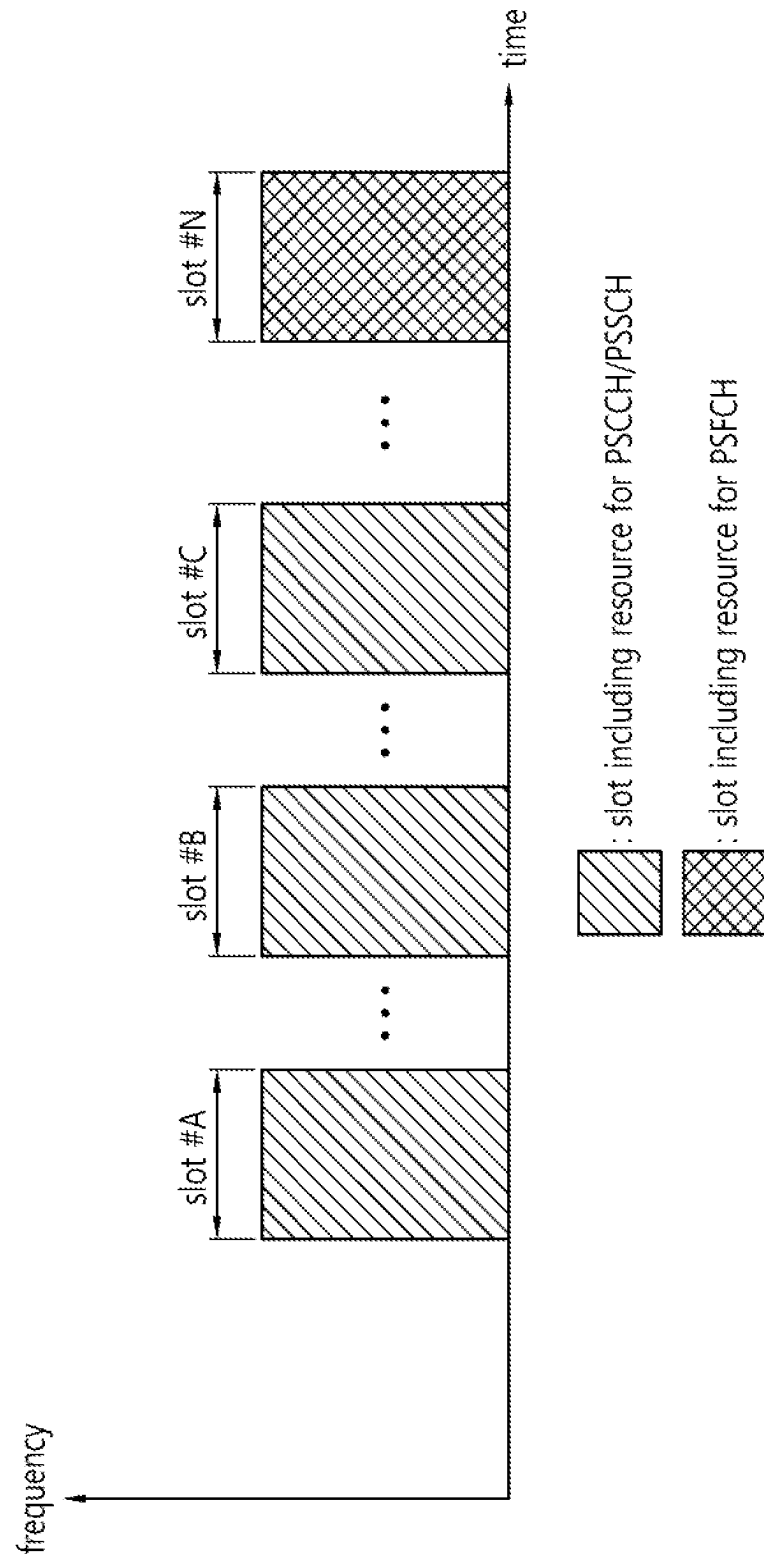


FIG. 17

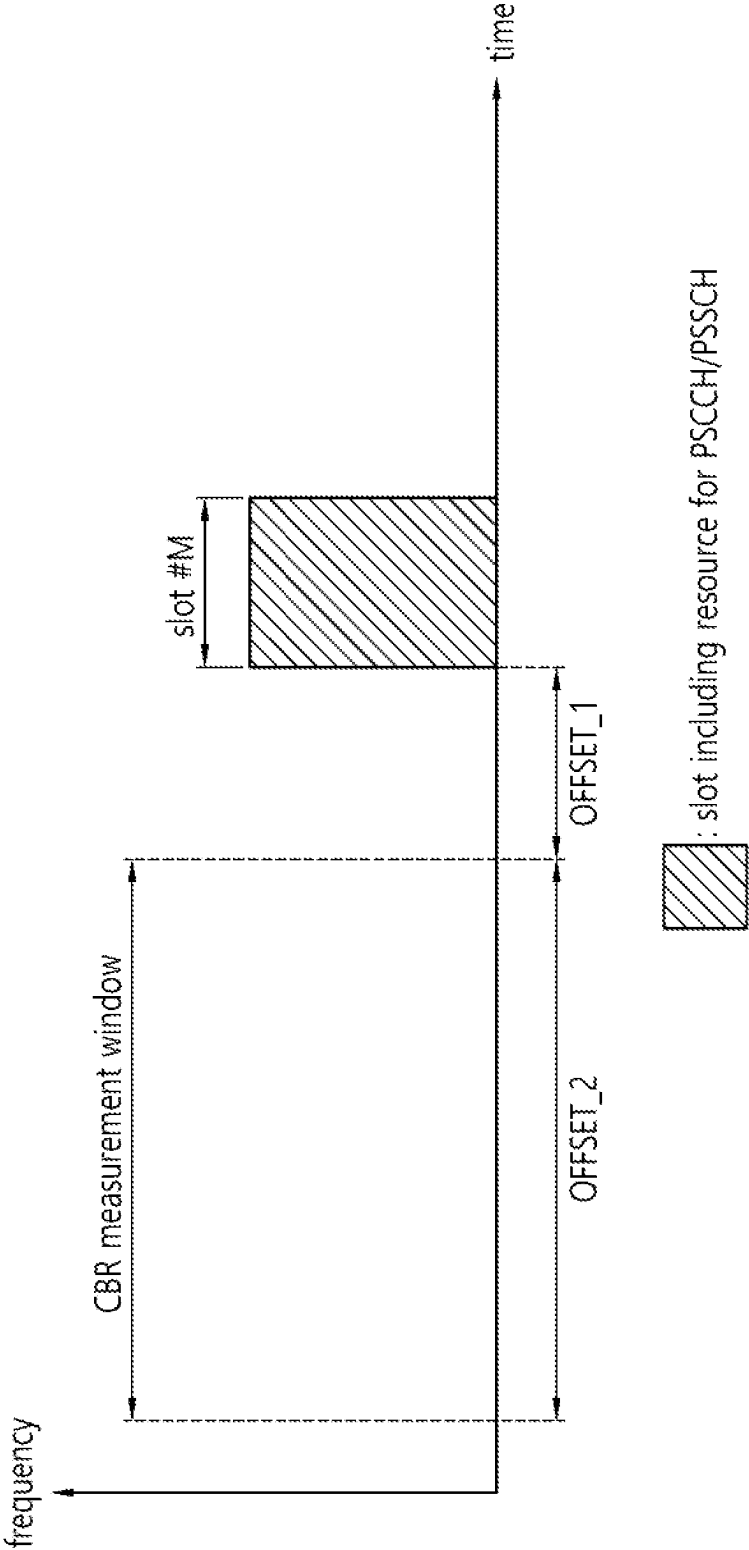


FIG. 18

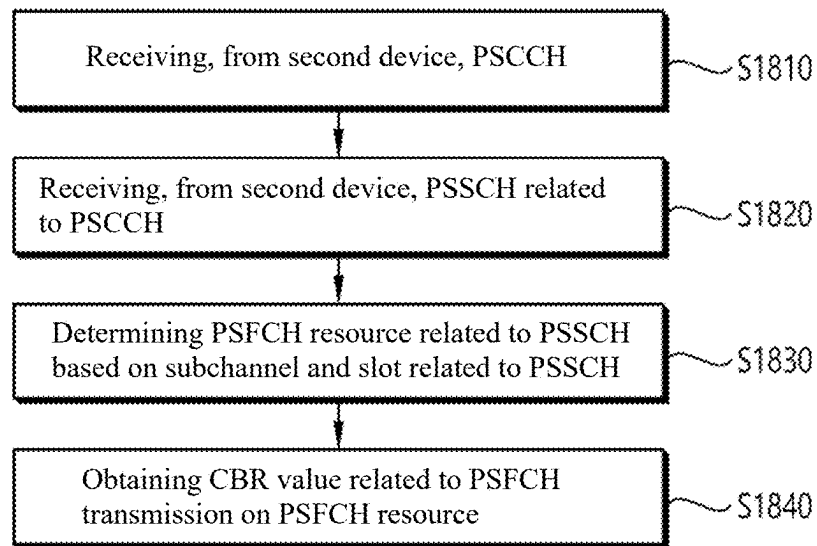


FIG. 19

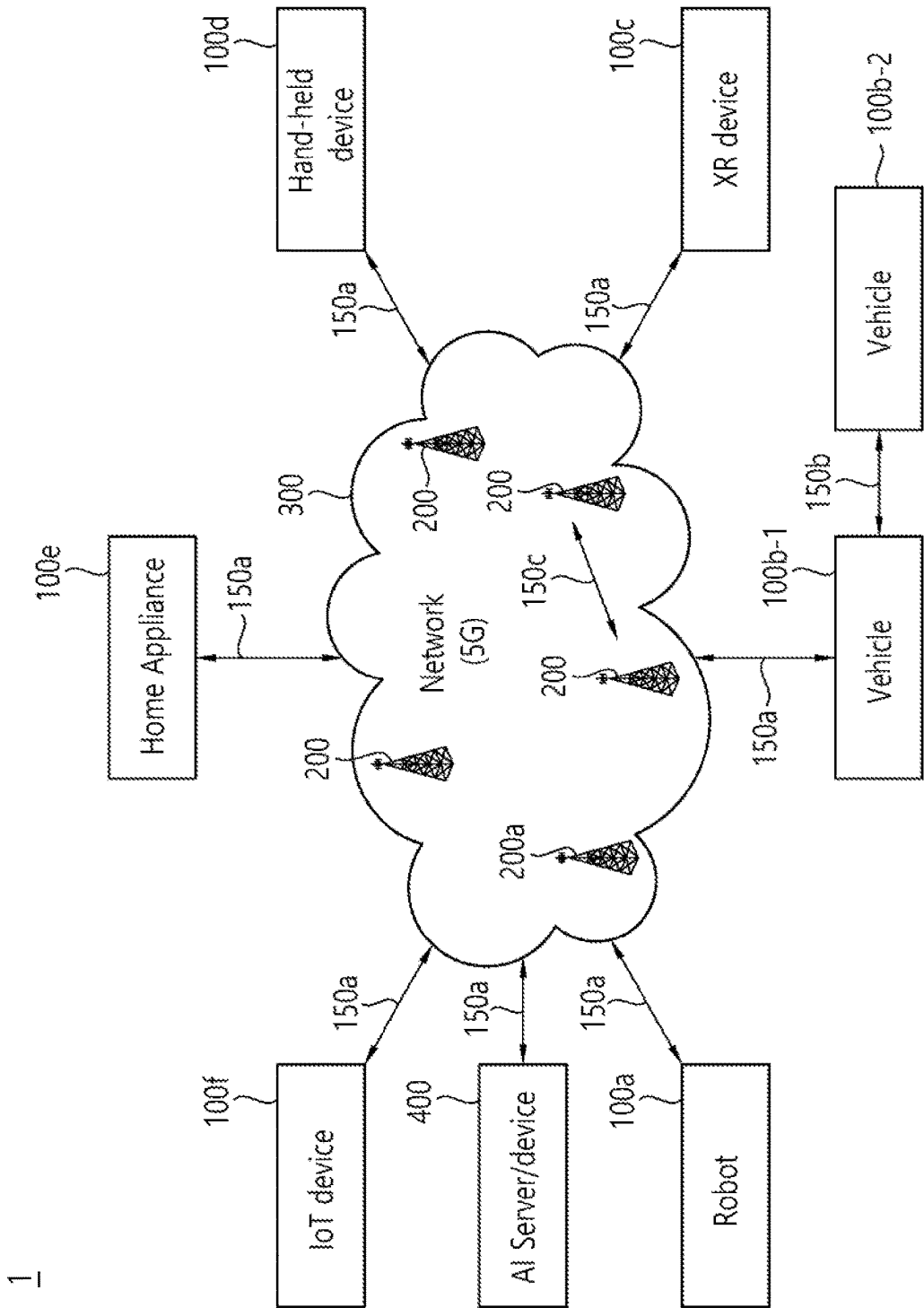


FIG. 20

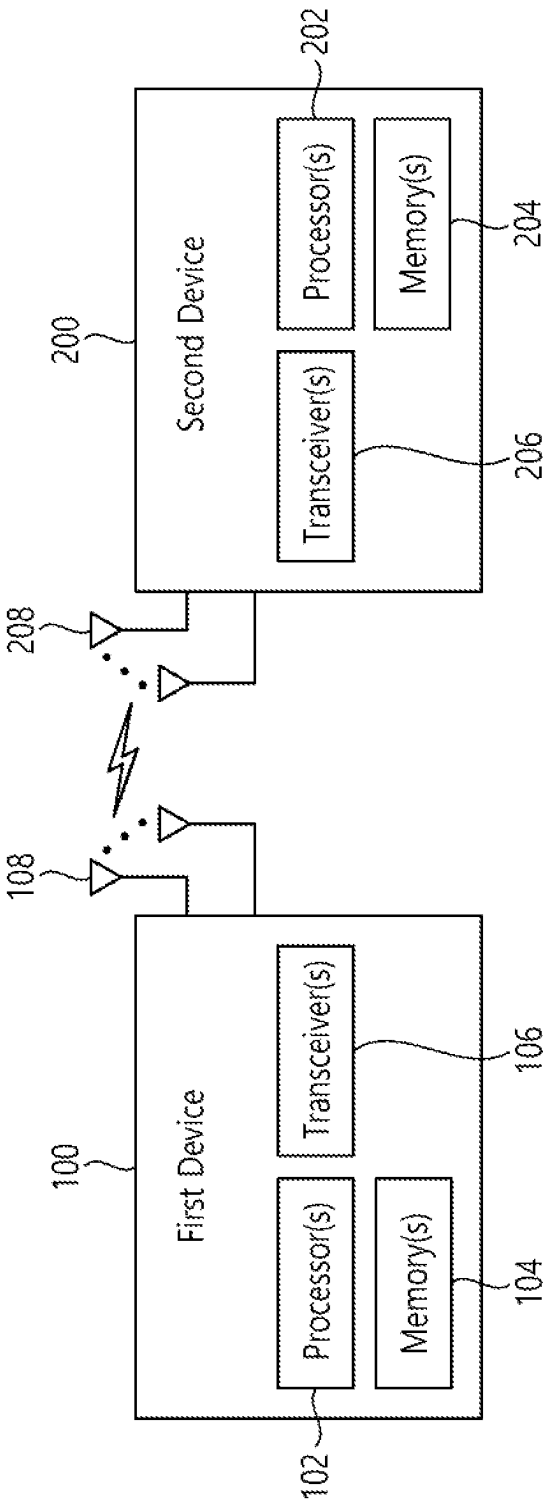


FIG. 21

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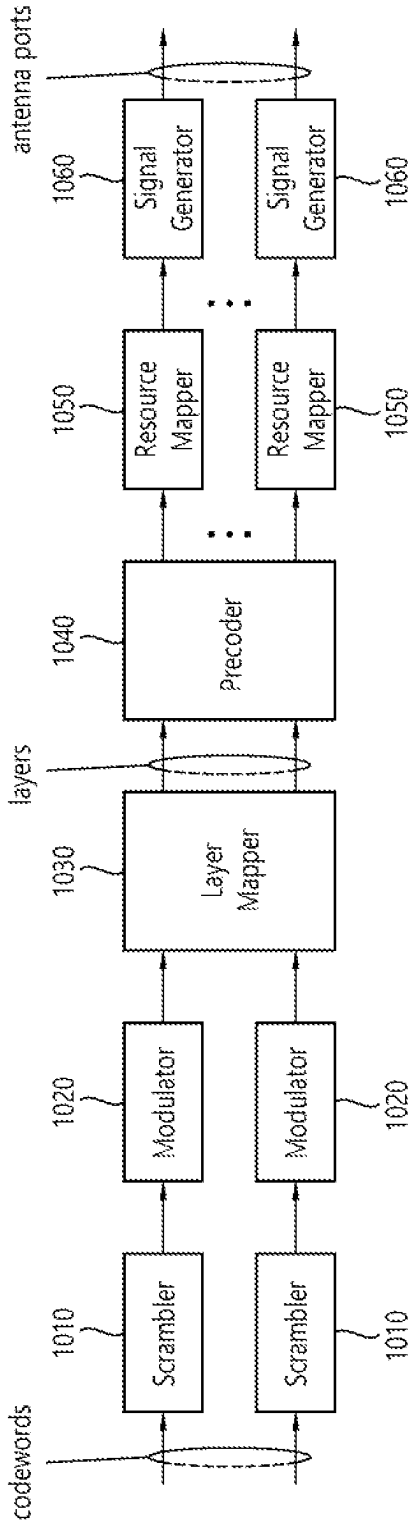


FIG. 22

Device (100,200)

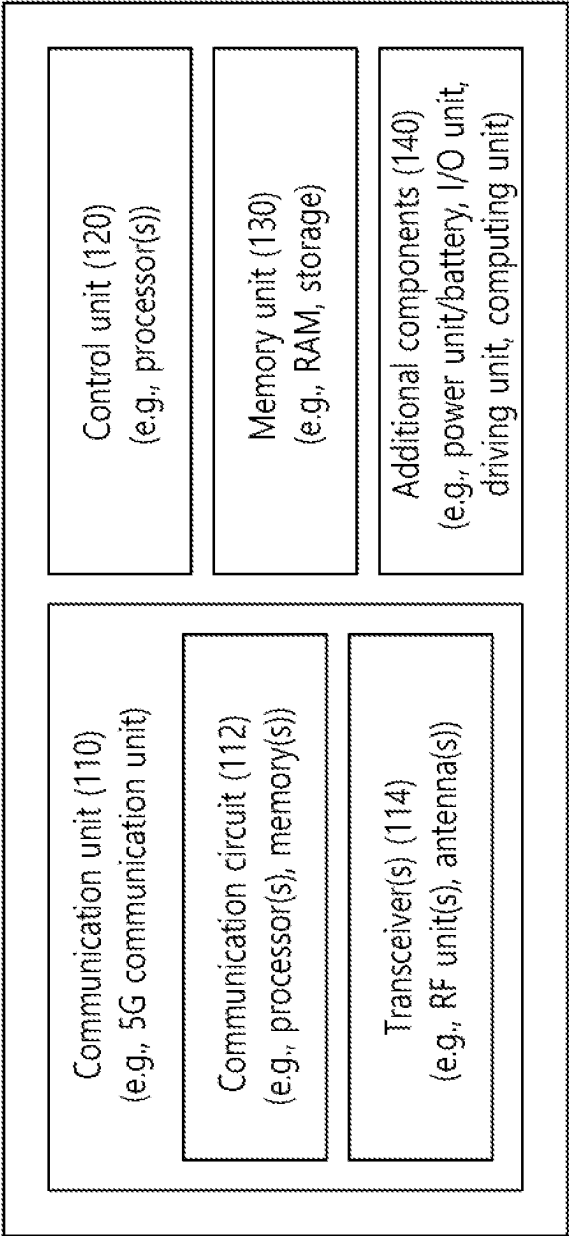




FIG. 23

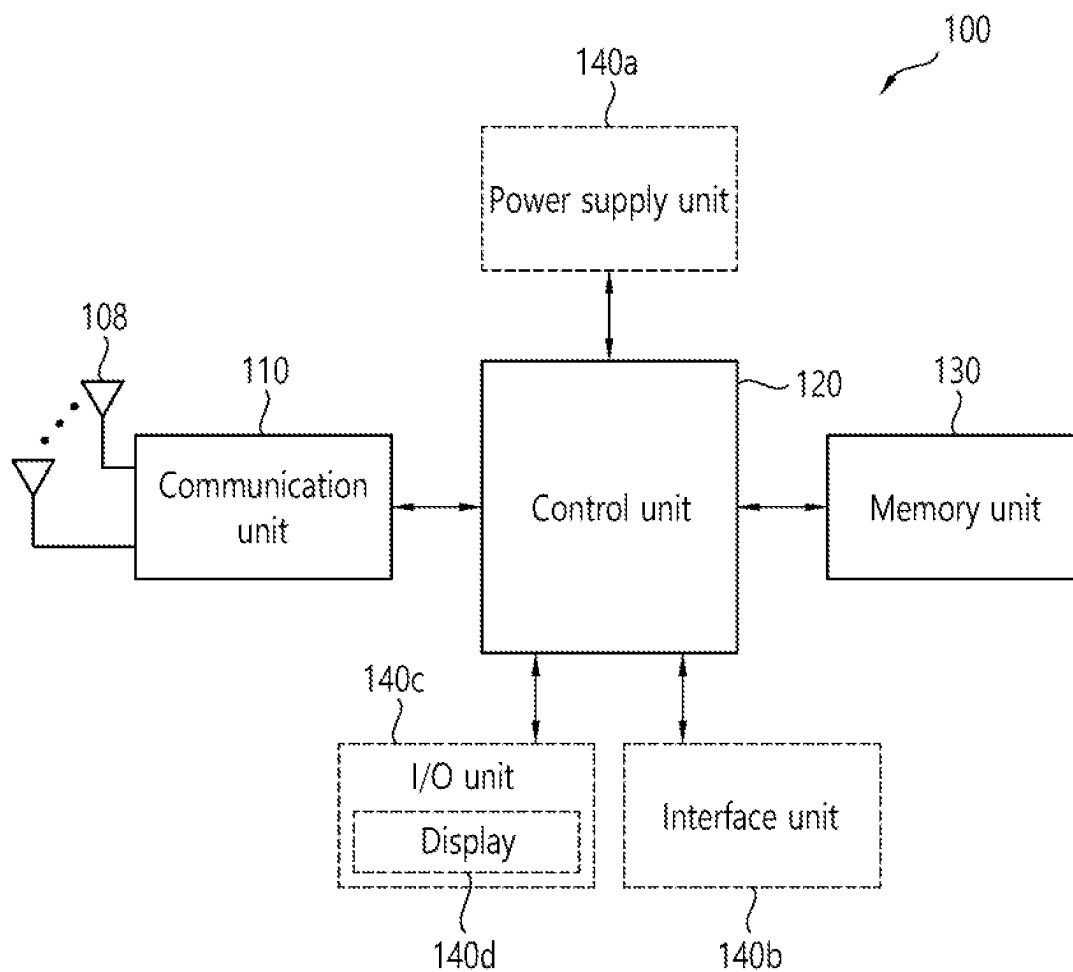
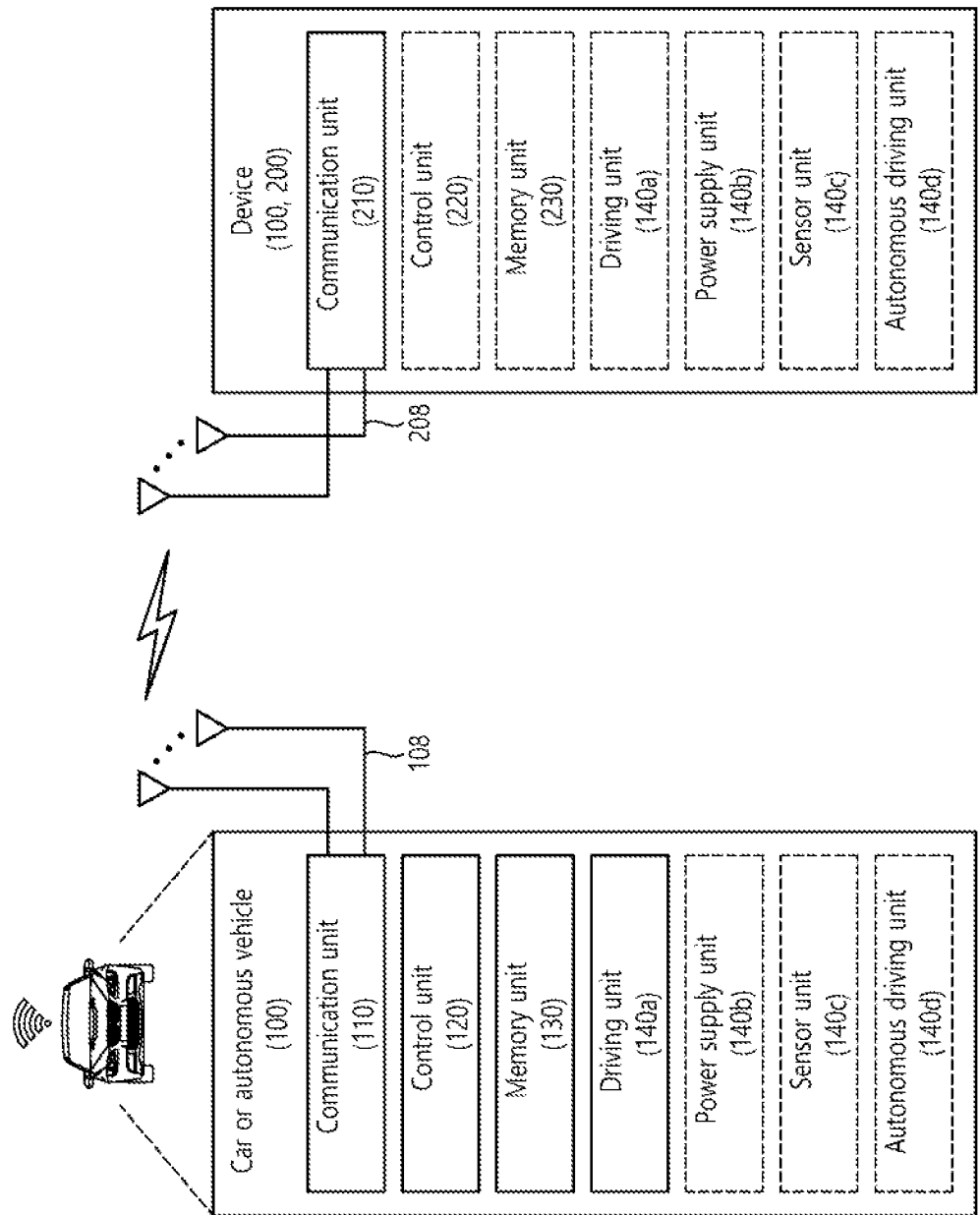


FIG. 24



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## METHOD AND APPARATUS FOR PERFORMING CONGESTION CONTROL IN NR V2X

### CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2021/001773, filed on Feb. 10, 2021, which claims the benefit of earlier filing date and right of priority to Korean Application No. 10-2020-0015897, filed on Feb. 10, 2020, the contents of which are all hereby incorporated by reference herein in their entireties.

### BACKGROUND OF THE DISCLOSURE

#### Field of the Disclosure

This disclosure relates to a wireless communication system.

#### Related Art

Sidelink (SL) communication is a communication scheme in which a direct link is established between User Equipments (UEs) and the UEs exchange voice and data directly with each other without intervention of an evolved Node B (eNB). SL communication is under consideration as a solution to the overhead of an eNB caused by rapidly increasing data traffic.

Vehicle-to-everything (V2X) refers to a communication technology through which a vehicle exchanges information with another vehicle, a pedestrian, an object having an infrastructure (or infra) established therein, and so on. The V2X may be divided into 4 types, such as vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P). The V2X communication may be provided via a PC5 interface and/or Uu interface.

Meanwhile, as a wider range of communication devices require larger communication capacities, the need for mobile broadband communication that is more enhanced than the existing Radio Access Technology (RAT) is rising. Accordingly, discussions are made on services and user equipment (UE) that are sensitive to reliability and latency. And, a next generation radio access technology that is based on the enhanced mobile broadband communication, massive Machine Type Communication (MTC), Ultra-Reliable and Low Latency Communication (URLLC), and so on, may be referred to as a new radio access technology (RAT) or new radio (NR). Herein, the NR may also support vehicle-to-everything (V2X) communication.

FIG. 1 is a drawing for describing V2X communication based on NR, compared to V2X communication based on RAT used before NR. The embodiment of FIG. 1 may be combined with various embodiments of the present disclosure.

Regarding V2X communication, a scheme of providing a safety service, based on a V2X message such as Basic Safety Message (BSM), Cooperative Awareness Message (CAM), and Decentralized Environmental Notification Message (DENM) is focused in the discussion on the RAT used before the NR. The V2X message may include position information, dynamic information, attribute information, or

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the like. For example, a UE may transmit a periodic message type CAM and/or an event triggered message type DENM to another UE.

For example, the CAM may include dynamic state information of the vehicle such as direction and speed, static data of the vehicle such as a size, and basic vehicle information such as an exterior illumination state, route details, or the like. For example, the UE may broadcast the CAM, and latency of the CAM may be less than 100 ms. For example, the UE may generate the DENM and transmit it to another UE in an unexpected situation such as a vehicle breakdown, accident, or the like. For example, all vehicles within a transmission range of the UE may receive the CAM and/or the DENM. In this case, the DENM may have a higher priority than the CAM.

Thereafter, regarding V2X communication, various V2X scenarios are proposed in NR. For example, the various V2X scenarios may include vehicle platooning, advanced driving, extended sensors, remote driving, or the like.

For example, based on the vehicle platooning, vehicles may move together by dynamically forming a group. For example, in order to perform platoon operations based on the vehicle platooning, the vehicles belonging to the group may receive periodic data from a leading vehicle. For example, the vehicles belonging to the group may decrease or increase an interval between the vehicles by using the periodic data.

For example, based on the advanced driving, the vehicle may be semi-automated or fully automated. For example, each vehicle may adjust trajectories or maneuvers, based on data obtained from a local sensor of a proximity vehicle and/or a proximity logical entity. In addition, for example, each vehicle may share driving intention with proximity vehicles.

For example, based on the extended sensors, raw data, processed data, or live video data obtained through the local sensors may be exchanged between a vehicle, a logical entity, a UE of pedestrians, and/or a V2X application server. Therefore, for example, the vehicle may recognize a more improved environment than an environment in which a self-sensor is used for detection.

For example, based on the remote driving, for a person who cannot drive or a remote vehicle in a dangerous environment, a remote driver or a V2X application may operate or control the remote vehicle. For example, if a route is predictable such as public transportation, cloud computing based driving may be used for the operation or control of the remote vehicle. In addition, for example, an access for a cloud-based back-end service platform may be considered for the remote driving.

Meanwhile, a scheme of specifying service requirements for various V2X scenarios such as vehicle platooning, advanced driving, extended sensors, remote driving, or the like is discussed in NR-based V2X communication.

### SUMMARY OF THE DISCLOSURE

#### Technical Objects

Meanwhile, in the conventional NR V2X, only PSCCH/PSSCH related CBR measurement and congestion control may be supported. For this reason, there may be a limit in adaptively guaranteeing the performance of the PSFCH having a performance requirement different from that of the PSCCH/PSSCH according to a change in the interference level on the (pre-configured) PSFCH resource domain. In addition, since there is also PSCCH/PSSCH that does not require SL HARQ feedback (through PSFCH) among

PSCCH/PSSCH transmitted on the (pre-configured) PSCCH/PSSCH resource domain, it may be difficult to consider that the interference level (e.g., CBR) measured in the PSCCH/PSSCH resource domain (accurately) represents the interference level of the PSFCH resource domain. Accordingly, there is a need to propose a method for a UE to perform congestion control and an apparatus supporting the same.

### Technical Solutions

In one embodiment, a method for performing, by a first device, wireless communication is provided. The method may comprise: receiving, from a second device, a physical sidelink control channel (PSCCH); receiving, from the second device, a physical sidelink shared channel (PSSCH) related to the PSCCH; determining a physical sidelink feedback channel (PSFCH) resource related to the PSSCH based on a subchannel and a slot related to the PSSCH; and obtaining a channel busy ratio (CBR) value related to PSFCH transmission on the PSFCH resource, wherein the CBR value is obtained based on measurement in a time interval between (i) a first slot before a first offset from a slot in which the PSSCH related to the PSFCH resource is received and (ii) a second slot before a second offset from the first slot.

In one embodiment, a first device configured to perform wireless communication is provided. The first device may comprise: one or more memories storing instructions; one or more transceivers; and one or more processors connected to the one or more memories and the one or more transceivers. The one or more processors may execute the instructions to: receive, from a second device, a physical sidelink control channel (PSCCH); receive, from the second device, a physical sidelink shared channel (PSSCH) related to the PSCCH; determine a physical sidelink feedback channel (PSFCH) resource related to the PSSCH based on a subchannel and a slot related to the PSSCH; and obtain a channel busy ratio (CBR) value related to PSFCH transmission on the PSFCH resource, wherein the CBR value is obtained based on measurement in a time interval between (i) a first slot before a first offset from a slot in which the PSSCH related to the PSFCH resource is received and (ii) a second slot before a second offset from the first slot.

### Effects of the Disclosure

The user equipment (UE) may efficiently perform SL communication.

### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a drawing for describing V2X communication based on NR, compared to V2X communication based on RAT used before NR.

FIG. 2 shows a structure of an NR system, based on an embodiment of the present disclosure.

FIG. 3 shows a functional division between an NG-RAN and a 5GC, based on an embodiment of the present disclosure.

FIG. 4 shows a radio protocol architecture, based on an embodiment of the present disclosure.

FIG. 5 shows a structure of an NR system, based on an embodiment of the present disclosure.

FIG. 6 shows a structure of a slot of an NR frame, based on an embodiment of the present disclosure.

FIG. 7 shows an example of a BWP, based on an embodiment of the present disclosure.

FIG. 8 shows a radio protocol architecture for a SL communication, based on an embodiment of the present disclosure.

FIG. 9 shows a UE performing V2X or SL communication, based on an embodiment of the present disclosure.

FIG. 10 shows a procedure of performing V2X or SL communication by a UE based on a transmission mode, based on an embodiment of the present disclosure.

FIG. 11 shows three cast types, based on an embodiment of the present disclosure.

FIG. 12 shows a resource unit for CBR measurement, based on an embodiment of the present disclosure.

FIG. 13 shows a method for a UE to determine a CBR measurement window, based on an embodiment of the present disclosure.

FIG. 14 shows a method for a UE to determine a CBR measurement window, based on an embodiment of the present disclosure.

FIG. 15 shows a method for a UE to determine a CBR measurement window, based on an embodiment of the present disclosure.

FIG. 16 shows a method in which a UE that has received a plurality of PSCCHs/PSSCHs transmits PSFCH, based on an embodiment of the present disclosure.

FIG. 17 shows a method for a UE to determine a CBR measurement window, based on an embodiment of the present disclosure.

FIG. 18 shows a method for a first device to perform wireless communication, based on an embodiment of the present disclosure.

FIG. 19 shows a communication system 1, based on an embodiment of the present disclosure.

FIG. 20 shows wireless devices, based on an embodiment of the present disclosure.

FIG. 21 shows a signal process circuit for a transmission signal, based on an embodiment of the present disclosure.

FIG. 22 shows another example of a wireless device, based on an embodiment of the present disclosure.

FIG. 23 shows a hand-held device, based on an embodiment of the present disclosure.

FIG. 24 shows a vehicle or an autonomous vehicle, based on an embodiment of the present disclosure.

### DESCRIPTION OF EXEMPLARY EMBODIMENTS

In the present disclosure, “A or B” may mean “only A”, “only B” or “both A and B.” In other words, in the present disclosure, “A or B” may be interpreted as “A and/or B”. For example, in the present disclosure, “A, B, or C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, C”.

A slash (/) or comma used in the present disclosure may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B, or C”.

In the present disclosure, “at least one of A and B” may mean “only A”, “only B”, or “both A and B”. In addition, in the present disclosure, the expression “at least one of A or B” or “at least one of A and/or B” may be interpreted as “at least one of A and B”.

In addition, in the present disclosure, “at least one of A, B, and C” may mean “only A”, “only B”, “only C”, or “any

combination of A, B, and C". In addition, "at least one of A, B, or C" or "at least one of A, B, and/or C" may mean "at least one of A, B, and C".

In addition, a parenthesis used in the present disclosure may mean "for example". Specifically, when indicated as "control information (PDCCH)", it may mean that "PDCCH" is proposed as an example of the "control information". In other words, the "control information" of the present disclosure is not limited to "PDCCH", and "PDCCH" may be proposed as an example of the "control information". In addition, when indicated as "control information (i.e., PDCCH)", it may also mean that "PDCCH" is proposed as an example of the "control information".

A technical feature described individually in one figure in the present disclosure may be individually implemented, or may be simultaneously implemented.

The technology described below may be used in various wireless communication systems such as code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), orthogonal frequency division multiple access (OFDMA), single carrier frequency division multiple access (SC-FDMA), and so on. The CDMA may be implemented with a radio technology, such as universal terrestrial radio access (UTRA) or CDMA-2000. The TDMA may be implemented with a radio technology, such as global system for mobile communications (GSM)/general packet radio service (GPRS)/enhanced data rate for GSM evolution (EDGE). The OFDMA may be implemented with a radio technology, such as institute of electrical and electronics engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, evolved UTRA (E-UTRA), and so on. IEEE 802.16m is an evolved version of IEEE 802.16e and provides backward compatibility with a system based on the IEEE 802.16e. The UTRA is part of a universal mobile telecommunication system (UMTS). 3rd generation partnership project (3GPP) long term evolution (LTE) is part of an evolved UMTS (E-UMTS) using the E-UTRA. The 3GPP LTE uses the OFDMA in a downlink and uses the SC-FDMA in an uplink. LTE-advanced (LTE-A) is an evolution of the LTE.

5G NR is a successive technology of LTE-A corresponding to a new Clean-slate type mobile communication system having the characteristics of high performance, low latency, high availability, and so on. 5G NR may use resources of all spectrum available for usage including low frequency bands of less than 1 GHz, middle frequency bands ranging from 1 GHz to 10 GHz, high frequency (millimeter waves) of 24 GHz or more, and so on.

For clarity in the description, the following description will mostly focus on LTE-A or 5G NR. However, technical features according to an embodiment of the present disclosure will not be limited only to this.

FIG. 2 shows a structure of an NR system, based on an embodiment of the present disclosure. The embodiment of FIG. 2 may be combined with various embodiments of the present disclosure.

Referring to FIG. 2, a next generation-radio access network (NG-RAN) may include a BS 20 providing a UE 10 with a user plane and control plane protocol termination. For example, the BS 20 may include a next generation-Node B (gNB) and/or an evolved-NodeB (eNB). For example, the UE 10 may be fixed or mobile and may be referred to as other terms, such as a mobile station (MS), a user terminal (UT), a subscriber station (SS), a mobile terminal (MT), wireless device, and so on. For example, the BS may be referred to as a fixed station which communicates with the

UE 10 and may be referred to as other terms, such as a base transceiver system (BTS), an access point (AP), and so on.

The embodiment of FIG. 2 exemplifies a case where only the gNB is included. The BSs 20 may be connected to one another via Xn interface. The BS 20 may be connected to one another via 5th generation (5G) core network (5GC) and NG interface. More specifically, the BSs 20 may be connected to an access and mobility management function (AMF) 30 via NG-C interface, and may be connected to a user plane function (UPF) 30 via NG-U interface.

FIG. 3 shows a functional division between an NG-RAN and a 5GC, based on an embodiment of the present disclosure. The embodiment of FIG. 3 may be combined with various embodiments of the present disclosure.

Referring to FIG. 3, the gNB may provide functions, such as Inter Cell Radio Resource Management (RRM), Radio Bearer (RB) control, Connection Mobility Control, Radio Admission Control, Measurement Configuration & Provision, Dynamic Resource Allocation, and so on. An AMF may provide functions, such as Non Access Stratum (NAS) security, idle state mobility processing, and so on. A UPF may provide functions, such as Mobility Anchoring, Protocol Data Unit (PDU) processing, and so on. A Session Management Function (SMF) may provide functions, such as user equipment (UE) Internet Protocol (IP) address allocation, PDU session control, and so on.

Layers of a radio interface protocol between the UE and the network can be classified into a first layer (L1), a second layer (L2), and a third layer (L3) based on the lower three layers of the open system interconnection (OSI) model that is well-known in the communication system. Among them, a physical (PHY) layer belonging to the first layer provides an information transfer service by using a physical channel, and a radio resource control (RRC) layer belonging to the third layer serves to control a radio resource between the UE and the network. For this, the RRC layer exchanges an RRC message between the UE and the BS.

FIG. 4 shows a radio protocol architecture, based on an embodiment of the present disclosure. The embodiment of FIG. 4 may be combined with various embodiments of the present disclosure. Specifically, FIG. 4(a) shows a radio protocol architecture for a user plane, and FIG. 4(b) shows a radio protocol architecture for a control plane. The user plane corresponds to a protocol stack for user data transmission, and the control plane corresponds to a protocol stack for control signal transmission.

Referring to FIG. 4, a physical layer provides an upper layer with an information transfer service through a physical channel. The physical layer is connected to a medium access control (MAC) layer which is an upper layer of the physical layer through a transport channel. Data is transferred between the MAC layer and the physical layer through the transport channel. The transport channel is classified according to how and with what characteristics data is transmitted through a radio interface.

Between different physical layers, i.e., a physical layer of a transmitter and a physical layer of a receiver, data are transferred through the physical channel. The physical channel is modulated using an orthogonal frequency division multiplexing (OFDM) scheme, and utilizes time and frequency as a radio resource.

The MAC layer provides services to a radio link control (RLC) layer, which is a higher layer of the MAC layer, via a logical channel. The MAC layer provides a function of mapping multiple logical channels to multiple transport channels. The MAC layer also provides a function of logical channel multiplexing by mapping multiple logical channels

to a single transport channel. The MAC layer provides data transfer services over logical channels.

The RLC layer performs concatenation, segmentation, and reassembly of Radio Link Control Service Data Unit (RLC SDU). In order to ensure diverse quality of service (QoS) required by a radio bearer (RB), the RLC layer provides three types of operation modes, i.e., a transparent mode (TM), an unacknowledged mode (UM), and an acknowledged mode (AM). An AM RLC provides error correction through an automatic repeat request (ARQ).

A radio resource control (RRC) layer is defined only in the control plane. The RRC layer serves to control the logical channel, the transport channel, and the physical channel in association with configuration, reconfiguration and release of RBs. The RB is a logical path provided by the first layer (i.e., the physical layer or the PHY layer) and the second layer (i.e., the MAC layer, the RLC layer, and the packet data convergence protocol (PDCP) layer) for data delivery between the UE and the network.

Functions of a packet data convergence protocol (PDCP) layer in the user plane include user data delivery, header compression, and ciphering. Functions of a PDCP layer in the control plane include control-plane data delivery and ciphering/integrity protection.

A service data adaptation protocol (SDAP) layer is defined only in a user plane. The SDAP layer performs mapping between a Quality of Service (QoS) flow and a data radio bearer (DRB) and QoS flow ID (QFI) marking in both DL and UL packets.

The configuration of the RB implies a process for specifying a radio protocol layer and channel properties to provide a particular service and for determining respective detailed parameters and operations. The RB can be classified into two types, i.e., a signaling RB (SRB) and a data RB (DRB). The SRB is used as a path for transmitting an RRC message in the control plane. The DRB is used as a path for transmitting user data in the user plane.

When an RRC connection is established between an RRC layer of the UE and an RRC layer of the E-UTRAN, the UE is in an RRC\_CONNECTED state, and, otherwise, the UE may be in an RRC\_IDLE state. In case of the NR, an RRC\_INACTIVE state is additionally defined, and a UE being in the RRC\_INACTIVE state may maintain its connection with a core network whereas its connection with the BS is released.

Data is transmitted from the network to the UE through a downlink transport channel. Examples of the downlink transport channel include a broadcast channel (BCH) for transmitting system information and a downlink-shared channel (SCH) for transmitting user traffic or control messages. Traffic of downlink multicast or broadcast services or the control messages can be transmitted on the downlink-SCH or an additional downlink multicast channel (MCH). Data is transmitted from the UE to the network through an uplink transport channel. Examples of the uplink transport channel include a random access channel (RACH) for transmitting an initial control message and an uplink SCH for transmitting user traffic or control messages.

Examples of logical channels belonging to a higher channel of the transport channel and mapped onto the transport channels include a broadcast channel (BCCH), a paging control channel (PCCH), a common control channel (CCCH), a multicast control channel (MCCH), a multicast traffic channel (MTCH), etc.

The physical channel includes several OFDM symbols in a time domain and several sub-carriers in a frequency domain. One sub-frame includes a plurality of OFDM

symbols in the time domain. A resource block is a unit of resource allocation, and consists of a plurality of OFDM symbols and a plurality of sub-carriers. Further, each sub-frame may use specific sub-carriers of specific OFDM symbols (e.g., a first OFDM symbol) of a corresponding subframe for a physical downlink control channel (PDCCH), i.e., an L1/L2 control channel. A transmission time interval (TTI) is a unit time of subframe transmission.

FIG. 5 shows a structure of an NR system, based on an embodiment of the present disclosure. The embodiment of FIG. 5 may be combined with various embodiments of the present disclosure.

Referring to FIG. 5, in the NR, a radio frame may be used for performing uplink and downlink transmission. A radio frame has a length of 10 ms and may be defined to be configured of two half-frames (HFs). A half-frame may include five 1 ms subframes (SFs). A subframe (SF) may be divided into one or more slots, and the number of slots within a subframe may be determined based on subcarrier spacing (SCS). Each slot may include 12 or 14 OFDM(A) symbols according to a cyclic prefix (CP).

In case of using a normal CP, each slot may include 14 symbols. In case of using an extended CP, each slot may include 12 symbols. Herein, a symbol may include an OFDM symbol (or CP-OFDM symbol) and a Single Carrier-FDMA (SC-FDMA) symbol (or Discrete Fourier Transform-spread-OFDM (DFT-s-OFDM) symbol).

Table 1 shown below represents an example of a number of symbols per slot ( $N_{slot}^{symb}$ ), a number slots per frame ( $N_{frame, u}^{slot}$ ), and a number of slots per subframe ( $N_{subframe, u}^{slot}$ ) based on an SCS configuration (u), in a case where a normal CP is used.

TABLE 1

| SCS ( $15 \cdot 2^u$ ) | $N_{slot}^{symb}$ | $N_{frame, u}^{slot}$ | $N_{subframe, u}^{slot}$ |
|------------------------|-------------------|-----------------------|--------------------------|
| 15 KHz (u = 0)         | 14                | 10                    | 1                        |
| 30 KHz (u = 1)         | 14                | 20                    | 2                        |
| 60 KHz (u = 2)         | 14                | 40                    | 4                        |
| 120 KHz (u = 3)        | 14                | 80                    | 8                        |
| 240 KHz (u = 4)        | 14                | 160                   | 16                       |

Table 2 shows an example of a number of symbols per slot, a number of slots per frame, and a number of slots per subframe based on the SCS, in a case where an extended CP is used.

TABLE 2

| SCS ( $15 \cdot 2^u$ ) | $N_{slot}^{symb}$ | $N_{frame, u}^{slot}$ | $N_{subframe, u}^{slot}$ |
|------------------------|-------------------|-----------------------|--------------------------|
| 60 KHz (u = 2)         | 12                | 40                    | 4                        |

In an NR system, OFDM(A) numerologies (e.g., SCS, CP length, and so on) between multiple cells being integrate to one UE may be differently configured. Accordingly, a (absolute time) duration (or section) of a time resource (e.g., subframe, slot or TTI) (collectively referred to as a time unit (TU) for simplicity) being configured of the same number of symbols may be differently configured in the integrated cells.

In the NR, multiple numerologies or SCSs for supporting diverse 5G services may be supported. For example, in case an SCS is 15 kHz, a wide area of the conventional cellular bands may be supported, and, in case an SCS is 30 kHz/60 kHz a dense-urban, lower latency, wider carrier bandwidth may be supported. In case the SCS is 60 kHz or higher, a

bandwidth that is greater than 24.25 GHz may be used in order to overcome phase noise.

An NR frequency band may be defined as two different types of frequency ranges. The two different types of frequency ranges may be FR1 and FR2. The values of the frequency ranges may be changed (or varied), and, for example, the two different types of frequency ranges may be as shown below in Table 3. Among the frequency ranges that are used in an NR system, FR1 may mean a “sub 6 GHz range”, and FR2 may mean an “above 6 GHz range” and may also be referred to as a millimeter wave (mmW).

TABLE 3

| Frequency Range designation | Corresponding frequency range | Subcarrier Spacing (SCS) |
|-----------------------------|-------------------------------|--------------------------|
| FR1                         | 450 MHz-6000 MHz              | 15, 30, 60 kHz           |
| FR2                         | 24250 MHz-52600 MHz           | 60, 120, 240 kHz         |

As described above, the values of the frequency ranges in the NR system may be changed (or varied). For example, as shown below in Table 4, FR1 may include a band within a range of 410 MHz to 7125 MHz. More specifically, FR1 may include a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, and so on) and higher. For example, a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, and so on) and higher being included in FR1 may include an unlicensed band. The unlicensed band may be used for diverse purposes, e.g., the unlicensed band for vehicle-specific communication (e.g., automated driving).

TABLE 4

| Frequency Range designation | Corresponding frequency range | Subcarrier Spacing (SCS) |
|-----------------------------|-------------------------------|--------------------------|
| FR1                         | 410 MHz-7125 MHz              | 15, 30, 60 kHz           |
| FR2                         | 24250 MHz-52600 MHz           | 60, 120, 240 kHz         |

FIG. 6 shows a structure of a slot of an NR frame, based on an embodiment of the present disclosure. The embodiment of FIG. 6 may be combined with various embodiments of the present disclosure.

Referring to FIG. 6, a slot includes a plurality of symbols in a time domain. For example, in case of a normal CP, one slot may include 14 symbols. However, in case of an extended CP, one slot may include 12 symbols. Alternatively, in case of a normal CP, one slot may include 7 symbols. However, in case of an extended CP, one slot may include 6 symbols.

A carrier includes a plurality of subcarriers in a frequency domain. A Resource Block (RB) may be defined as a plurality of consecutive subcarriers (e.g., 12 subcarriers) in the frequency domain. A Bandwidth Part (BWP) may be defined as a plurality of consecutive (Physical) Resource Blocks ((P)RBs) in the frequency domain, and the BWP may correspond to one numerology (e.g., SCS, CP length, and so on). A carrier may include a maximum of N number BWPs (e.g., 5 BWPs). Data communication may be performed via an activated BWP. Each element may be referred to as a Resource Element (RE) within a resource grid and one complex symbol may be mapped to each element.

Meanwhile, a radio interface between a UE and another UE or a radio interface between the UE and a network may consist of an L1 layer, an L2 layer, and an L3 layer. In various embodiments of the present disclosure, the L1 layer may imply a physical layer. In addition, for example, the L2

layer may imply at least one of a MAC layer, an RLC layer, a PDCP layer, and an SDAP layer. In addition, for example, the L3 layer may imply an RRC layer.

Hereinafter, a bandwidth part (BWP) and a carrier will be described.

The BWP may be a set of consecutive physical resource blocks (PRBs) in a given numerology. The PRB may be selected from consecutive sub-sets of common resource blocks (CRBs) for the given numerology on a given carrier.

When using bandwidth adaptation (BA), a reception bandwidth and transmission bandwidth of a UE are not necessarily as large as a bandwidth of a cell, and the reception bandwidth and transmission bandwidth of the BS may be adjusted. For example, a network/BS may inform the UE of bandwidth adjustment. For example, the UE receive information/configuration for bandwidth adjustment from the network/BS. In this case, the UE may perform bandwidth adjustment based on the received information/configuration. For example, the bandwidth adjustment may include an increase/decrease of the bandwidth, a position change of the bandwidth, or a change in subcarrier spacing of the bandwidth.

For example, the bandwidth may be decreased during a period in which activity is low to save power. For example, the position of the bandwidth may move in a frequency domain. For example, the position of the bandwidth may move in the frequency domain to increase scheduling flexibility. For example, the subcarrier spacing of the bandwidth may be changed. For example, the subcarrier spacing of the bandwidth may be changed to allow a different service. A subset of a total cell bandwidth of a cell may be called a bandwidth part (BWP). The BA may be performed when the BS/network configures the BWP to the UE and the BS/network informs the UE of the BWP currently in an active state among the configured BWPs.

For example, the BWP may be at least any one of an active BWP, an initial BWP, and/or a default BWP. For example, the UE may not monitor downlink radio link quality in a DL BWP other than an active DL BWP on a primary cell (PCell). For example, the UE may not receive PDCCH, physical downlink shared channel (PDSCH), or channel state information-reference signal (CSI-RS) (excluding RRM) outside the active DL BWP. For example, the UE may not trigger a channel state information (CSI) report for the inactive DL BWP. For example, the UE may not transmit physical uplink control channel (PUCCH) or physical uplink shared channel (PUSCH) outside an active UL BWP. For example, in a downlink case, the initial BWP may be given as a consecutive RB set for a remaining minimum system information (RMSI) control resource set (CORE-SET) (configured by physical broadcast channel (PBCH)). For example, in an uplink case, the initial BWP may be given by system information block (SIB) for a random access procedure. For example, the default BWP may be configured by a higher layer. For example, an initial value of the default BWP may be an initial DL BWP. For energy saving, if the UE fails to detect downlink control information (DCI) during a specific period, the UE may switch the active BWP of the UE to the default BWP.

Meanwhile, the BWP may be defined for SL. The same SL BWP may be used in transmission and reception. For example, a transmitting UE may transmit a SL channel or a SL signal on a specific BWP, and a receiving UE may receive the SL channel or the SL signal on the specific BWP. In a licensed carrier, the SL BWP may be defined separately from a Uu BWP, and the SL BWP may have configuration signaling separate from the Uu BWP. For example, the UE

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may receive a configuration for the SL BWP from the BS/network. The SL BWP may be (pre-)configured in a carrier with respect to an out-of-coverage NR V2X UE and an RRC\_IDLE UE. For the UE in the RRC\_CONNECTED mode, at least one SL BWP may be activated in the carrier.

FIG. 7 shows an example of a BWP, based on an embodiment of the present disclosure. The embodiment of FIG. 7 may be combined with various embodiments of the present disclosure. It is assumed in the embodiment of FIG. 7 that the number of BWPs is 3.

Referring to FIG. 7, a common resource block (CRB) may be a carrier resource block numbered from one end of a carrier band to the other end thereof. In addition, the PRB may be a resource block numbered within each BWP. A point A may indicate a common reference point for a resource block grid.

The BWP may be configured by a point A, an offset  $N_{BWP}^{start}$  from the point A, and a bandwidth  $N_{BWP}^{size}$ . For example, the point A may be an external reference point of a PRB of a carrier in which a subcarrier 0 of all numerologies (e.g., all numerologies supported by a network on that carrier) is aligned. For example, the offset may be a PRB interval between a lowest subcarrier and the point A in a given numerology. For example, the bandwidth may be the number of PRBs in the given numerology.

Hereinafter, V2X or SL communication will be described.

FIG. 8 shows a radio protocol architecture for a SL communication, based on an embodiment of the present disclosure. The embodiment of FIG. 8 may be combined with various embodiments of the present disclosure. More specifically, FIG. 8(a) shows a user plane protocol stack, and FIG. 8(b) shows a control plane protocol stack.

Hereinafter, a sidelink synchronization signal (SLSS) and synchronization information will be described.

The SLSS may include a primary sidelink synchronization signal (PSSS) and a secondary sidelink synchronization signal (SSSS), as a SL-specific sequence. The PSSS may be referred to as a sidelink primary synchronization signal (S-PSS), and the SSSS may be referred to as a sidelink secondary synchronization signal (S-SSS). For example, length-127 M-sequences may be used for the S-PSS, and length-127 gold sequences may be used for the S-SSS. For example, a UE may use the S-PSS for initial signal detection and for synchronization acquisition. For example, the UE may use the S-PSS and the S-SSS for acquisition of detailed synchronization and for detection of a synchronization signal ID.

A physical sidelink broadcast channel (PSBCH) may be a (broadcast) channel for transmitting default (system) information which must be first known by the UE before SL signal transmission/reception. For example, the default information may be information related to SLSS, a duplex mode (DM), a time division duplex (TDD) uplink/downlink (UL/DL) configuration, information related to a resource pool, a type of an application related to the SLSS, a subframe offset, broadcast information, or the like. For example, for evaluation of PSBCH performance, in NR V2X, a payload size of the PSBCH may be 56 bits including 24-bit cyclic redundancy check (CRC).

The S-PSS, the S-SSS, and the PSBCH may be included in a block format (e.g., SL synchronization signal (SS)/PSBCH block, hereinafter, sidelink-synchronization signal block (S-SSB)) supporting periodical transmission. The S-SSB may have the same numerology (i.e., SCS and CP length) as a physical sidelink control channel (PSCCH)/physical sidelink shared channel (PSSCH) in a carrier, and a transmission bandwidth may exist within a (pre-)config-

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ured sidelink (SL) BWP. For example, the S-SSB may have a bandwidth of 11 resource blocks (RBs). For example, the PSBCH may exist across 11 RBs. In addition, a frequency position of the S-SSB may be (pre-)configured. Accordingly, the UE does not have to perform hypothesis detection at frequency to discover the S-SSB in the carrier.

FIG. 9 shows a UE performing V2X or SL communication, based on an embodiment of the present disclosure. The embodiment of FIG. 9 may be combined with various embodiments of the present disclosure.

Referring to FIG. 9, in V2X or SL communication, the term 'UE' may generally imply a UE of a user. However, if a network equipment such as a BS transmits/receives a signal according to a communication scheme between UEs, the BS may also be regarded as a sort of the UE. For example, a UE 1 may be a first apparatus 100, and a UE 2 may be a second apparatus 200.

For example, the UE 1 may select a resource unit corresponding to a specific resource in a resource pool which implies a set of series of resources. In addition, the UE 1 may transmit a SL signal by using the resource unit. For example, a resource pool in which the UE 1 is capable of transmitting a signal may be configured to the UE 2 which is a receiving UE, and the signal of the UE 1 may be detected in the resource pool.

Herein, if the UE 1 is within a connectivity range of the BS, the BS may inform the UE 1 of the resource pool. Otherwise, if the UE 1 is out of the connectivity range of the BS, another UE may inform the UE 1 of the resource pool, or the UE 1 may use a pre-configured resource pool.

In general, the resource pool may be configured in unit of a plurality of resources, and each UE may select a unit of one or a plurality of resources to use it in SL signal transmission thereof.

Hereinafter, resource allocation in SL will be described.

FIG. 10 shows a procedure of performing V2X or SL communication by a UE based on a transmission mode, based on an embodiment of the present disclosure. The embodiment of FIG. 10 may be combined with various embodiments of the present disclosure. In various embodiments of the present disclosure, the transmission mode may be called a mode or a resource allocation mode. Hereinafter, for convenience of explanation, in LTE, the transmission mode may be called an LTE transmission mode. In NR, the transmission mode may be called an NR resource allocation mode.

For example, FIG. 10(a) shows a UE operation related to an LTE transmission mode 1 or an LTE transmission mode 3. Alternatively, for example, FIG. 10(a) shows a UE operation related to an NR resource allocation mode 1. For example, the LTE transmission mode 1 may be applied to general SL communication, and the LTE transmission mode 3 may be applied to V2X communication.

For example, FIG. 10(b) shows a UE operation related to an LTE transmission mode 2 or an LTE transmission mode 4. Alternatively, for example, FIG. 10(b) shows a UE operation related to an NR resource allocation mode 2.

Referring to FIG. 10(a), in the LTE transmission mode 1, the LTE transmission mode 3, or the NR resource allocation mode 1, a BS may schedule a SL resource to be used by the UE for SL transmission. For example, the BS may perform resource scheduling to a UE 1 through a PDCCH (more specifically, downlink control information (DCI)), and the UE 1 may perform V2X or SL communication with respect to a UE 2 according to the resource scheduling. For example, the UE 1 may transmit a sidelink control information (SCI) to the UE 2 through a physical sidelink control channel



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(PSCCH), and thereafter transmit data based on the SCI to the UE 2 through a physical sidelink shared channel (PSSCH).

Referring to FIG. 10(b), in the LTE transmission mode 2, the LTE transmission mode 4, or the NR resource allocation mode 2, the UE may determine a SL transmission resource within a SL resource configured by a BS/network or a pre-configured SL resource. For example, the configured SL resource or the pre-configured SL resource may be a resource pool. For example, the UE may autonomously select or schedule a resource for SL transmission. For example, the UE may perform SL communication by autonomously selecting a resource within a configured resource pool. For example, the UE may autonomously select a resource within a selective window by performing a sensing and resource (re)selection procedure. For example, the sensing may be performed in unit of subchannels. In addition, the UE 1 which has autonomously selected the resource within the resource pool may transmit the SCI to the UE 2 through a PSCCH, and thereafter may transmit data based on the SCI to the UE 2 through a PSSCH.

FIG. 11 shows three cast types, based on an embodiment of the present disclosure. The embodiment of FIG. 11 may be combined with various embodiments of the present disclosure. Specifically, FIG. 11(a) shows broadcast-type SL communication, FIG. 11(b) shows unicast type-SL communication, and FIG. 11(c) shows groupcast-type SL communication. In case of the unicast-type SL communication, a UE may perform one-to-one communication with respect to another UE. In case of the groupcast-type SL transmission, the UE may perform SL communication with respect to one or more UEs in a group to which the UE belongs. In various embodiments of the present disclosure, SL groupcast communication may be replaced with SL multicast communication, SL one-to-many communication, or the like.

Hereinafter, sidelink (SL) congestion control will be described.

If a UE autonomously determines an SL transmission resource, the UE also autonomously determines a size and frequency of use for a resource used by the UE. Of course, due to a constraint from a network or the like, it may be restricted to use a resource size or frequency of use, which is greater than or equal to a specific level. However, if all UEs use a relatively great amount of resources in a situation where many UEs are concentrated in a specific region at a specific time, overall performance may significantly deteriorate due to mutual interference.

Accordingly, the UE may need to observe a channel situation. If it is determined that an excessively great amount of resources are consumed, it is preferable that the UE autonomously decreases the use of resources. In the present disclosure, this may be defined as congestion control (CR). For example, the UE may determine whether energy measured in a unit time/frequency resource is greater than or equal to a specific level, and may adjust an amount and frequency of use for its transmission resource based on a ratio of the unit time/frequency resource in which the energy greater than or equal to the specific level is observed. In the present disclosure, the ratio of the time/frequency resource in which the energy greater than or equal to the specific level is observed may be defined as a channel busy ratio (CBR). The UE may measure the CBR for a channel/frequency. Additionally, the UE may transmit the measured CBR to the network/BS.

FIG. 12 shows a resource unit for CBR measurement, based on an embodiment of the present disclosure. The

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embodiment of FIG. 12 may be combined with various embodiments of the present disclosure.

Referring to FIG. 12, CBR may denote the number of sub-channels in which a measurement result value of a received signal strength indicator (RSSI) has a value greater than or equal to a pre-configured threshold as a result of measuring the RSSI by a UE on a sub-channel basis for a specific period (e.g., 100 ms). Alternatively, the CBR may denote a ratio of sub-channels having a value greater than or equal to a pre-configured threshold among sub-channels for a specific duration. For example, in the embodiment of FIG. 12, if it is assumed that a hatched sub-channel is a sub-channel having a value greater than or equal to a pre-configured threshold, the CBR may denote a ratio of the hatched sub-channels for a period of 100 ms. Additionally, the UE may report the CBR to the BS.

Further, congestion control considering a priority of traffic (e.g., packet) may be necessary. To this end, for example, the UE may measure a channel occupancy ratio (CR). Specifically, the UE may measure the CBR, and the UE may determine a maximum value  $CRLimit_k$  of a channel occupancy ratio  $k$  (CR $_k$ ) that can be occupied by traffic corresponding to each priority (e.g.,  $k$ ) based on the CBR. For example, the UE may derive the maximum value  $CRLimit_k$  of the channel occupancy ratio with respect to a priority of each traffic, based on a predetermined table of CBR measurement values. For example, in case of traffic having a relatively high priority, the UE may derive a maximum value of a relatively great channel occupancy ratio. Thereafter, the UE may perform congestion control by restricting a total sum of channel occupancy ratios of traffic, of which a priority  $k$  is lower than  $i$ , to a value less than or equal to a specific value. Based on this method, the channel occupancy ratio may be more strictly restricted for traffic having a relatively low priority.

In addition thereto, the UE may perform SL congestion control by using a method of adjusting a level of transmit power, dropping a packet, determining whether retransmission is to be performed, adjusting a transmission RB size (Modulation and Coding Scheme (MCS) coordination), or the like.

When UEs transmit signals, if the number of surrounding UEs is excessively big (or an amount of measured interference is excessively large), each individual UE changes a transmission power, a size of a frequency resource in use, a size of a time resource in use, a packet transmission periodicity and the like, thereby reducing influence on the surrounding UEs. In doing so, the UEs may determine different transmission parameters depending on significances (or priorities) of transmitted packets. Thus, a UE's operation of recognizing a surrounding situation and changing a transmission parameter will be called a congestion control. Regarding such a congestion control, a UE to which several resource regions (e.g., a resource pool) are allocated may perform a separate congestion control in each pool.

For the congestion control, a UE may measure a Channel Busy Ratio (CBR) for surrounding situation recognition. The CBR may be defined as a ratio of resources determined as used by another UE in comparison to total resources. Whether each resource is used may be determined by the number of subchannels having RSSI exceeding a predetermined threshold over the total number of subchannels in a predetermined time interval based on whether a reception power or RSSI measured on the corresponding resource exceeds the predetermined threshold. Channel occupancy ratio (CR) evaluated at subframe  $n$  may be defined as below. The CR may mean the total number of sub-channels used for

its transmissions in subframes  $[n-a, n-1]$  and granted in subframes  $[n, n+b]$  divided by the total number of configured sub-channels in the transmission pool over  $[n-a, n+b]$ . Channel busy ratio (CBR) measured in subframe  $n$  may be defined as follows. CBR may mean, for PSSCH, the portion of sub-channels in the resource pool whose S-RSSI measured by the UE exceed a pre-configured threshold sensed over subframes  $[n-100, n-1]$ . CBR may mean, for PSCCH, in a pool pre-configured such that PSCCH may be transmitted with its corresponding PSSCH in non-adjacent resource blocks, the portion of the resources of the PSCCH pool whose S-RSSI measured by the UE exceed a pre-configured threshold sensed over subframes  $[n-100, n-1]$ , assuming that the PSCCH pool is composed of resources with a size of two consecutive PRB pairs in the frequency domain. The CBR may be applied to RRC\_IDLE intra-frequency, RRC\_IDLE inter-frequency, RRC\_CONNECTED intra-frequency and RRC\_CONNECTED inter-frequency. Here, the subframe index may be based on physical subframe index.

A UE measures a CBR within a resource region and changes all or some of CR (channel ratio or resource utilization, i.e., a ratio of a resource used by each UE to total resources), MCS, RB size (subchannel size, where a sub-channel means a size of a basic frequency resource of which resource allocation occurs on a UE-to-UE communication occurring band), retransmission number, transmission power, resource reservation interval and the like for the measured CBR each. The CR may be applied to RRC\_IDLE intra-frequency, RRC\_IDLE inter-frequency, RRC\_CONNECTED intra-frequency and RRC\_CONNECTED inter-frequency. Here,  $a$  may be a positive integer and  $b$  may be 0 or a positive integer.  $a$  and  $b$  may be determined by UE implementation with ' $a+b+1=1000$ ', ' $a \geq 500$ ', and ' $n+b$  should not exceed the last transmission opportunity of the grant for the current transmission'. Here, CR may be evaluated for each (re)transmission. Here, in evaluating CR, the UE may assume the transmission parameter used at subframe  $n$  is reused according to the existing grant(s) in subframes  $[n+1, n+b]$  without packet dropping. Here, the subframe index may be based on physical subframe index. Here, the CR may be computed per priority level.

Hereinafter, a hybrid automatic repeat request (HARQ) procedure will be described.

An error compensation scheme is used to secure communication reliability. Examples of the error compensation scheme may include a forward error correction (FEC) scheme and an automatic repeat request (ARQ) scheme. In the FEC scheme, errors in a receiving end are corrected by attaching an extra error correction code to information bits. The FEC scheme has an advantage in that time delay is small and no information is additionally exchanged between a transmitting end and the receiving end but also has a disadvantage in that system efficiency deteriorates in a good channel environment. The ARQ scheme has an advantage in that transmission reliability can be increased but also has a disadvantage in that a time delay occurs and system efficiency deteriorates in a poor channel environment.

A hybrid automatic repeat request (HARQ) scheme is a combination of the FEC scheme and the ARQ scheme. In the HARQ scheme, it is determined whether an unrecoverable error is included in data received by a physical layer, and retransmission is requested upon detecting the error, thereby improving performance.

In case of SL unicast and groupcast, HARQ feedback and HARQ combining in the physical layer may be supported. For example, when a receiving UE operates in a resource

allocation mode 1 or 2, the receiving UE may receive the PSSCH from a transmitting UE, and the receiving UE may transmit HARQ feedback for the PSSCH to the transmitting UE by using a sidelink feedback control information (SFCI) format through a physical sidelink feedback channel (PSFCH).

For example, the SL HARQ feedback may be enabled for unicast. In this case, in a non-code block group (non-CBG) operation, if the receiving UE decodes a PSCCH of which a target is the receiving UE and if the receiving UE successfully decodes a transport block related to the PSCCH, the receiving UE may generate HARQ-ACK. In addition, the receiving UE may transmit the HARQ-ACK to the transmitting UE. Otherwise, if the receiving UE cannot successfully decode the transport block after decoding the PSCCH of which the target is the receiving UE, the receiving UE may generate the HARQ-NACK. In addition, the receiving UE may transmit HARQ-NACK to the transmitting UE.

For example, the SL HARQ feedback may be enabled for groupcast. For example, in the non-CBG operation, two HARQ feedback options may be supported for groupcast.

- (1) Groupcast option 1: After the receiving UE decodes the PSCCH of which the target is the receiving UE, if the receiving UE fails in decoding of a transport block related to the PSCCH, the receiving UE may transmit HARQ-NACK to the transmitting UE through a PSFCH. Otherwise, if the receiving UE decodes the PSCCH of which the target is the receiving UE and if the receiving UE successfully decodes the transport block related to the PSCCH, the receiving UE may not transmit the HARQ-ACK to the transmitting UE.
- (2) Groupcast option 2: After the receiving UE decodes the PSCCH of which the target is the receiving UE, if the receiving UE fails in decoding of the transport block related to the PSCCH, the receiving UE may transmit HARQ-NACK to the transmitting UE through the PSFCH. In addition, if the receiving UE decodes the PSCCH of which the target is the receiving UE and if the receiving UE successfully decodes the transport block related to the PSCCH, the receiving UE may transmit the HARQ-ACK to the transmitting UE through the PSFCH.

For example, if the groupcast option 1 is used in the SL HARQ feedback, all UEs performing groupcast communication may share a PSFCH resource. For example, UEs belonging to the same group may transmit HARQ feedback by using the same PSFCH resource.

For example, if the groupcast option 2 is used in the SL HARQ feedback, each UE performing groupcast communication may use a different PSFCH resource for HARQ feedback transmission. For example, UEs belonging to the same group may transmit HARQ feedback by using different PSFCH resources.

For example, when the SL HARQ feedback is enabled for groupcast, the receiving UE may determine whether to transmit the HARQ feedback to the transmitting UE based on a transmission-reception (TX-RX) distance and/or RSRP.

For example, in the groupcast option 1, in case of the TX-RX distance-based HARQ feedback, if the TX-RX distance is less than or equal to a communication range requirement, the receiving UE may transmit HARQ feedback for the PSSCH to the transmitting UE. Otherwise, if the TX-RX distance is greater than the communication range requirement, the receiving UE may not transmit the HARQ feedback for the PSSCH to the transmitting UE. For example, the transmitting UE may inform the receiving UE

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of a location of the transmitting UE through SCI related to the PSSCH. For example, the SCI related to the PSSCH may be second SCI. For example, the receiving UE may estimate or obtain the TX-RX distance based on a location of the receiving UE and the location of the transmitting UE. For example, the receiving UE may decode the SCI related to the PSSCH and thus may know the communication range requirement used in the PSSCH.

For example, in case of the resource allocation mode 1, a time (offset) between the PSFCH and the PSSCH may be configured or pre-configured. In case of unicast and groupcast, if retransmission is necessary on SL, this may be indicated to a BS by an in-coverage UE which uses the PUCCH. The transmitting UE may transmit an indication to a serving BS of the transmitting UE in a form of scheduling request (SR)/buffer status report (BSR), not a form of HARQ ACK/NACK. In addition, even if the BS does not receive the indication, the BS may schedule an SL retransmission resource to the UE. For example, in case of the resource allocation mode 2, a time (offset) between the PSFCH and the PSSCH may be configured or pre-configured.

For example, from a perspective of UE transmission in a carrier, TDM between the PSCCH/PSSCH and the PSFCH may be allowed for a PSFCH format for SL in a slot. For example, a sequence-based PSFCH format having a single symbol may be supported. Herein, the single symbol may not an automatic gain control (AGC) duration. For example, the sequence-based PSFCH format may be applied to unicast and groupcast.

For example, in a slot related to a resource pool, a PSFCH resource may be configured periodically as N slot durations, or may be pre-configured. For example, N may be configured as one or more values greater than or equal to 1. For example, N may be 1, 2, or 4. For example, HARQ feedback for transmission in a specific resource pool may be transmitted only through a PSFCH on the specific resource pool.

For example, if the transmitting UE transmits the PSSCH to the receiving UE across a slot #X to a slot #N, the receiving UE may transmit HARQ feedback for the PSSCH to the transmitting UE in a slot #(N+A). For example, the slot #(N+A) may include a PSFCH resource. Herein, for example, A may be a smallest integer greater than or equal to K. For example, K may be the number of logical slots. In this case, K may be the number of slots in a resource pool. Alternatively, for example, K may be the number of physical slots. In this case, K may be the number of slots inside or outside the resource pool.

For example, if the receiving UE transmits HARQ feedback on a PSFCH resource in response to one PSSCH transmitted by the transmitting UE to the receiving UE, the receiving UE may determine a frequency domain and/or code domain of the PSFCH resource based on an implicit mechanism in a configured resource pool. For example, the receiving UE may determine the frequency domain and/or code domain of the PSFCH resource, based on at least one of a slot index related to PSCCH/PSSCH/PSFCH, a sub-channel related to PSCCH/PSSCH, and/or an identifier for identifying each receiving UE in a group for HARQ feedback based on the groupcast option 2. Additionally/alternatively, for example, the receiving UE may determine the frequency domain and/or code domain of the PSFCH resource, based on at least one of SL RSRP, SINR, L1 source ID, and/or location information.

For example, if HARQ feedback transmission through the PSFCH of the UE and HARQ feedback reception through the PSFCH overlap, the UE may select any one of HARQ

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feedback transmission through the PSFCH and HARQ feedback reception through the PSFCH based on a priority rule. For example, the priority rule may be based on at least priority indication of the related PSCCH/PSSCH.

For example, if HARQ feedback transmission of a UE through a PSFCH for a plurality of UEs overlaps, the UE may select specific HARQ feedback transmission based on the priority rule. For example, the priority rule may be based on at least priority indication of the related PSCCH/PSSCH.

Meanwhile, in the present disclosure, for example, the term “configure/configured” or the term “define/defined” may refer to (pre)configuration from a base station or a network (through pre-defined signaling (e.g., SIB, MAC, RRC, etc.)) (for each resource pool). For example, “that A is configured” may mean “that the base station/network transmits information related to A to the UE”.

Meanwhile, in the LTE sidelink communication environment, if a plurality of resources are used by a plurality of UEs for a certain time, in order to adjust the corresponding problem, congestion control for PSCCH or PSSCH may be applied based on a channel occupancy ratio (CR) value or a channel busy ratio (CBR) value. For example, the UE may perform congestion control for the PSCCH or the PSSCH based on the CR value or the CBR value. Meanwhile, in the NR sidelink communication environment, PSFCH is newly introduced. Herein, in the case of congestion control for the PSFCH, the UE may use congestion control information for PSCCH or PSSCH related to the PSFCH.

In the conventional NR V2X, only PSCCH/PSSCH related CBR measurement and congestion control may be supported. For this reason, there may be a limit in adaptively guaranteeing the performance of the PSFCH having a performance requirement different from that of the PSCCH/PSSCH according to a change in the interference level on the (pre-configured) PSFCH resource domain. In addition, since there is also PSCCH/PSSCH that does not require SL HARQ feedback (through PSFCH) among PSCCH/PSSCH transmitted on the (pre-configured) PSCCH/PSSCH resource domain, it may be difficult to consider that the interference level (e.g., CBR) measured in the PSCCH/PSSCH resource domain (accurately) represents the interference level of the PSFCH resource domain.

Based on various embodiments of the present disclosure, a method for performing congestion control and an apparatus supporting the same are proposed. The method proposed according to various embodiments of the present disclosure may be extended to not only congestion control for PSFCH but also congestion control for PSCCH/PSSCH.

Based on an embodiment of the present disclosure, differently from congestion control for the PSCCH/PSSCH, congestion control for the PSFCH may be limitedly configured/applied only to a specific (pre-configured) transmission parameter. For example, only congestion control based on maximum transmission power limitation may be defined/allowed for the PSFCH. In this case, for example, the UE performing congestion control for the PSFCH may determine PSFCH transmit power based on a CR value or a CBR value related to the PSFCH. On the other hand, for example, when the UE performing congestion control for the PSFCH determines a parameter other than the PSFCH transmit power, the UE may not consider/use a CR value or a CBR value related to the PSFCH. Herein, for the following reasons, congestion control for the PSFCH may be limitedly configured/applied only to the specific (pre-configured) transmission parameter.

For example, the UE may determine whether to transmit the PSFCH based on PSCCH/PSSCH reception and/or an

indication of enabling SL HARQ feedback included in SCI. In this case, for example, if CR\_LIMIT-based congestion control is (already) applied to PSCCH/PSSCH transmission, CR\_LIMIT-based congestion control is implicitly applied to PSFCH transmission related to the PSCCH/PSSCH transmission. For the above reasons, unlike congestion control for the PSCCH/PSSCH, congestion control for the PSFCH may be limitedly configured/applied only to the specific (pre-configured) transmission parameter.

Alternatively, for example, the same as congestion control for the PSCCH/PSSCH, congestion control for the PSFCH may be configured/applied for (all) transmission parameters.

For example, the UE may configure/determine a CBR measurement period based on the rules proposed below. For example, if congestion control based on a specific pre-configured transmission parameter (e.g., maximum transmission power limitation) is applied to PSFCH transmission, the CBR measurement period used by the UE may be defined according to the following rule. In the present disclosure, for convenience of description, it may be assumed that PSFCH transmission is performed on slot #N, and it may be assumed that a priority of the PSFCH may be a priority of the associated PSSCH (e.g., data).

FIG. 13 or FIG. 14 shows a method for a UE to determine a CBR measurement window, based on an embodiment of the present disclosure. The embodiment of FIG. 13 or FIG. 14 may be combined with various embodiments of the present disclosure.

Referring to FIG. 13 or FIG. 14, the UE may use a CBR value measured in the period from slot # (N-OFFSET\_1) to slot # (N-OFFSET\_1-OFFSET\_2) for congestion control related to PSFCH transmission on slot #N. For example, the UE may perform CBR measurement in the time interval between slot # (N-OFFSET\_1) and slot # (N-OFFSET\_1-OFFSET\_2), and the UE may perform congestion control related to PSFCH transmission on slot #N based on the measured CBR value. In the embodiment of FIG. 13 or FIG. 14, the slot #N may be a slot including a PSFCH resource (e.g., a PSFCH symbol), and the slot # (N-K) may be a slot including a PSSCH/PSCCH resource (e.g., a PSSCH/PSCCH symbol) related to the PSFCH resource.

For example, the OFFSET\_1 value may be interpreted as a (minimum) time required to process the CBR measurement value and/or a (minimum) time required to set (PSFCH) transmit power. For example, the OFFSET\_1 value may be fixed to 4 ms. For example, the OFFSET\_1 value may be designated as a specific value to the UE by being configured or pre-configured from the network/base station. For example, the OFFSET\_1 value may be configured or pre-configured to 4 ms or 4 slots for the UE. For example, the OFFSET\_2 value may be a parameter that determines the length of the time window for CBR measurement. For example, the OFFSET\_2 value may be configured or pre-configured to 100 ms or 100 slots for the UE.

For example, the OFFSET\_1 value may be independently configured for the UE between the PSFCH and the PSCCH/PSSCH. For example, the OFFSET\_2 value may be independently configured for the UE between the PSFCH and the PSCCH/PSSCH. For example, between the PSFCH and the PSCCH/PSSCH, CBR measurement and congestion control related parameters (e.g., the length of the CBR measurement window, physical layer parameter limit for each CBR/packet priority, etc.) may be configured differently or independently for the UE.

For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a numerology (e.g., sub-carrier spacing, CP). For example, the

OFFSET\_1 value may be configured differently or independently for the UE based on a frequency range. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a service type. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a service priority. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a QoS requirement (e.g., reliability, latency). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a cast type (e.g., unicast, groupcast, broadcast). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a traffic type (e.g., a periodically generated packet or a non-periodically generated packet).

For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a numerology (e.g., sub-carrier spacing, CP). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a frequency range. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a service type. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a service priority. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a QoS requirement (e.g., reliability, latency). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a cast type (e.g., unicast, groupcast, broadcast). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a traffic type (e.g., a periodically generated packet or a non-periodically generated packet).

For example, the UE may repeatedly transmit PSFCH for a specific SL HARQ feedback through a plurality of resources (e.g., a plurality of slots or a plurality of symbols). In this case, for example, the UE may consider/determine a time when the first PSFCH transmission is performed as the slot #N. Alternatively, for example, the UE may consider/determine a time when the last PSFCH transmission is performed as the slot #N. Alternatively, for example, the UE may consider/determine a time when PSFCH transmission in a pre-configured order is performed as the slot #N. Alternatively, for example, the UE may consider/determine each time when PSFCH transmission is performed as the slot #N. In the above-described case, for example, the UE may use the interference value measured based on the PSFCH transmission time on the slot #N for the remaining repetitive PSFCH transmission related congestion control.

For example, the UE may repeatedly transmit PSCCH/PSSCH for a specific TB through a plurality of resources (e.g., a plurality of slots or a plurality of symbols). In this case, for example, the UE may consider/determine a time when the first PSCCH/PSSCH transmission is performed as the slot #N. Alternatively, for example, the UE may consider/determine a time when the last PSCCH/PSSCH transmission is performed as the slot #N. Alternatively, for example, the UE may consider/determine a time when PSCCH/PSSCH transmission in a pre-configured order is performed as the slot #N. Alternatively, for example, the UE may consider/determine each time when PSCCH/PSSCH transmission is performed as the slot #N.

FIG. 15 shows a method for a UE to determine a CBR measurement window, based on an embodiment of the present disclosure. The embodiment of FIG. 15 may be combined with various embodiments of the present disclosure.

Referring to FIG. 15, it is assumed that a PSCCH/PSSCH reception time related to PSFCH transmission is slot # (N-K). In this case, the UE may use a CBR value measured in the period from slot # (N-K-OFFSET\_1) to slot # (N-K-OFFSET\_1-OFFSET\_2) for congestion control related to PSFCH transmission on slot #N. For example, the UE may perform CBR measurement in the time interval between slot # (N-K-OFFSET\_1) and slot # (N-K-OFFSET\_1-OFFSET\_2), and the UE may perform congestion control related to PSFCH transmission on slot #N based on the measured CBR value. In the embodiment of FIG. 15, the slot #N may be a slot including a PSFCH resource (e.g., a PSFCH symbol), and the slot # (N-K) may be a slot including a PSSCH/PSCCH resource (e.g., a PSSCH/PSCCH symbol) related to the PSFCH resource.

For example, the OFFSET\_1 value may be interpreted as a (minimum) time required to process the CBR measurement value and/or a (minimum) time required to set (PSFCH) transmit power. For example, the OFFSET\_1 value may be fixed to 4 ms. For example, the OFFSET\_1 value may be designated as a specific value to the UE by being configured or pre-configured from the network/base station. For example, the OFFSET\_1 value may be configured or pre-configured to 4 ms or 4 slots for the UE. For example, the OFFSET\_2 value may be a parameter that determines the length of the time window for CBR measurement. For example, the OFFSET\_2 value may be configured or pre-configured to 100 ms or 100 slots for the UE.

For example, the OFFSET\_1 value may be independently configured for the UE between the PSFCH and the PSCCH/PSSCH. For example, the OFFSET\_2 value may be independently configured for the UE between the PSFCH and the PSCCH/PSSCH. For example, between the PSFCH and the PSCCH/PSSCH, CBR measurement and congestion control related parameters (e.g., the length of the CBR measurement window, physical layer parameter limit for each CBR/packet priority, etc.) may be configured differently or independently for the UE.

For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a numerology (e.g., sub-carrier spacing, CP). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a frequency range. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a service type. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a service priority. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a QoS requirement (e.g., reliability, latency). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a cast type (e.g., unicast, groupcast, broadcast). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a traffic type (e.g., a periodically generated packet or a non-periodically generated packet).

For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a numerology (e.g., sub-carrier spacing, CP). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a frequency range. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a service type. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a service priority. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a QoS

requirement (e.g., reliability, latency). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a cast type (e.g., unicast, groupcast, broadcast). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a traffic type (e.g., a periodically generated packet or a non-periodically generated packet).

FIG. 16 shows a method in which a UE that has received a plurality of PSCCHs/PSSCHs transmits PSFCH, based on an embodiment of the present disclosure. The embodiment of FIG. 16 may be combined with various embodiments of the present disclosure.

In the embodiment of FIG. 16, PSFCH transmitted in slot #N may be related to slot #A, slot #B, and slot #C. That is, the UE may perform PSFCH transmission in slot #N in response to a plurality of PSCCH/PSSCHs received in slot #A, slot #B, and slot #C. For example, the UE may receive PSCCH/PSSCH for a specific TB through a plurality of resources (e.g., a plurality of slots or a plurality of symbols), and the UE may perform PSFCH transmission related to the PSCCH/PSSCH. In this case, for example, the UE may consider/determine the time of the resource in which the last PSCCH/PSSCH transmission is performed (e.g., slot #C) as the slot # (N-K). Alternatively, for example, the UE may consider/determine the time of the resource in which the first PSCCH/PSSCH transmission is performed (e.g., slot #A) as the slot # (N-K). Alternatively, for example, the UE may consider/determine the time of the resource in which PSCCH/PSSCH transmission in a pre-configured order is performed as the slot # (N-K).

In the embodiment of FIG. 16, it is assumed that the UE fails to decode PSCCH transmitted on the slot #A. On the other hand, it is assumed that the UE succeeds in decoding PSCCH transmitted on the slot #B and the slot #C. In this case, for example, the UE may consider/determine the time of the last resource in which decoding for the PSCCH is actually successful (e.g., slot #C) as the slot # (N-K). Alternatively, for example, the UE may consider/determine the time of the first resource in which decoding for the PSCCH is actually successful (e.g., slot #B) as the slot # (N-K).

Based on an embodiment of the present disclosure, the UE may determine a CBR measurement window (e.g., a window in which interference is measured) for congestion control related to PSFCH transmitted on a PSFCH slot (i.e., slot #N). Herein, for example, a resource for which interference is measured within the CBR measurement window (hereinafter, CBR\_RSC) may be limited to a PSFCH resource domain. In this case, for example, the UE may obtain a CBR value, by measuring interference only in the PSFCH resource domain within the CBR measurement window. Or, for example, CBR\_RSC may be limited to the PSSCH/PSCCH resource domain. In this case, for example, the UE may obtain a CBR value, by measuring interference only in the PSSCH/PSCCH resource domain within the CBR measurement window. Or, for example, CBR\_RSC may include both the PSSCH/PSCCH resource domain and the PSFCH resource domain. In this case, for example, the UE may obtain a CBR value, by measuring interference in the PSSCH/PSCCH resource domain and the PSFCH resource domain within the CBR measurement window.

The (some) proposed method/principle (e.g., the embodiment related to FIG. 13 or FIG. 14) according to various embodiments of the present disclosure may be extended in order for the UE to determine/derive a CBR measurement period for congestion control for PSCCH/PSSCH.

FIG. 17 shows a method for a UE to determine a CBR measurement window, based on an embodiment of the present disclosure. The embodiment of FIG. 17 may be combined with various embodiments of the present disclosure.

Referring to FIG. 17, the UE may use a CBR value measured in the period from slot # (M-OFFSET\_1) to slot # (M-OFFSET\_1-OFFSET\_2) for congestion control related to PSCCH/PSSCH transmission on slot #M. For example, the UE may perform CBR measurement in the time interval between slot # (M-OFFSET\_1) and slot # (M-OFFSET\_1-OFFSET\_2), and the UE may perform congestion control related to PSCCH/PSSCH transmission on slot #M based on the measured CBR value. In the embodiment of FIG. 17, the slot #M may be a slot including a PSSCH/PSCCH resource (e.g., a PSSCH/PSCCH symbol).

For example, the OFFSET\_1 value may be interpreted as a (minimum) time required to process the CBR measurement value and/or a (minimum) time required to set (PSCCH/PSSCH) transmit power. For example, the OFFSET\_1 value may be fixed to 4 ms. For example, the OFFSET\_1 value may be designated as a specific value to the UE by being configured or pre-configured from the network/base station. For example, the OFFSET\_1 value may be configured or pre-configured to 4 ms or 4 slots for the UE. For example, the OFFSET\_2 value may be a parameter that determines the length of the time window for CBR measurement. For example, the OFFSET\_2 value may be configured or pre-configured to 100 ms or 100 slots for the UE.

For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a numerology (e.g., sub-carrier spacing, CP). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a frequency range. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a service type. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a service priority. For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a QoS requirement (e.g., reliability, latency). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a cast type (e.g., unicast, groupcast, broadcast). For example, the OFFSET\_1 value may be configured differently or independently for the UE based on a traffic type (e.g., a periodically generated packet or a non-periodically generated packet).

For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a numerology (e.g., sub-carrier spacing, CP). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a frequency range. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a service type. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a service priority. For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a QoS requirement (e.g., reliability, latency). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a cast type (e.g., unicast, groupcast, broadcast). For example, the OFFSET\_2 value may be configured differently or independently for the UE based on a traffic type (e.g., a periodically generated packet or a non-periodically generated packet).

For example, the UE may repeatedly transmit PSCCH/PSSCH for a specific TB through a plurality of resources (e.g., a plurality of slots or a plurality of symbols). In this case, for example, the UE may consider/determine a time when the first PSCCH/PSSCH transmission is performed as the slot #M. Alternatively, for example, the UE may consider/determine a time when the last PSCCH/PSSCH transmission is performed as the slot #M. Alternatively, for example, the UE may consider/determine a time when PSCCH/PSSCH transmission in a pre-configured order is performed as the slot #M. Alternatively, for example, the UE may consider/determine each time when PSCCH/PSSCH transmission is performed as the slot #M.

Based on various embodiments of the present disclosure, according to a change in interference in the PSFCH resource domain, the UE may efficiently perform PSFCH-related congestion control. Accordingly, the relevant performance requirements can be effectively satisfied.

FIG. 18 shows a method for a first device to perform wireless communication, based on an embodiment of the present disclosure. The embodiment of FIG. 18 may be combined with various embodiments of the present disclosure.

Referring to FIG. 18, in step S1810, the first device may receive, from a second device, a physical sidelink control channel (PSCCH). In step S1820, the first device may receive, from the second device, a physical sidelink shared channel (PSSCH) related to the PSCCH. In step S1830, the first device may determine a physical sidelink feedback channel (PSFCH) resource related to the PSSCH based on a subchannel and a slot related to the PSSCH. In step S1840, the first device may obtain a channel busy ratio (CBR) value related to PSFCH transmission on the PSFCH resource. For example, the CBR value may be obtained based on measurement in a time interval between (i) a first slot before a first offset from a slot in which the PSSCH related to the PSFCH resource is received and (ii) a second slot before a second offset from the first slot.

For example, based on a plurality of PSSCHs related to the PSFCH resource being received in a plurality of slots, the slot in which the PSSCH is received may be a last slot among the plurality of slots. For example, based on a plurality of PSSCHs related to the PSFCH resource being received in a plurality of slots, the slot in which the PSSCH is received may be the 1st slot among the plurality of slots. For example, based on a plurality of PSSCHs related to the PSFCH resource being received in a plurality of slots, the slot in which the PSSCH is received may be a slot in a pre-configured order for the first device among the plurality of slots.

For example, based on (i) a plurality of PSCCHs related to the PSFCH resource being transmitted in a plurality of slots, and (ii) successful decoding of at least one PSCCH in at least one slot among the plurality of slots by the first device, the slot in which the PSSCH is received may be a last slot among the at least one slot related to the at least one PSCCH. For example, based on (i) a plurality of PSCCHs related to the PSFCH resource being transmitted in a plurality of slots, and (ii) successful decoding of at least one PSCCH in at least one slot among the plurality of slots by the first device, the slot in which the PSSCH is received may be the 1st slot among the at least one slot related to the at least one PSCCH.

For example, the first offset may be an offset related to a time required for processing of the CBR value or an offset related to a time required for power setting for the PSFCH

transmission, and the second offset may be an offset for determining a length of a time window related to CBR measurement.

For example, the first offset or the second offset for obtaining the CBR value related to the PSFCH transmission and an offset for obtaining a CBR value related to PSSCH transmission or PSSCH transmission may be independently configured for the first device, respectively. For example, the first offset or the second offset may be configured differently based on at least one of a numerology, a frequency range, a service type, a service priority, a quality of service (QoS) requirement, a cast type, or a traffic type.

For example, the CBR value may be obtained based on the measurement only for PSFCH resource domain in the time interval between the first slot and the second slot. For example, the CBR value may be obtained based on the measurement only for PSSCH resource domain in the time interval between the first slot and the second slot. For example, the CBR value may be obtained based on the measurement for PSSCH resource domain, PSSCH resource domain, and PSFCH resource domain in the time interval between the first slot and the second slot.

Additionally, for example, the first device may determine a parameter related to the PSFCH transmission based on the CBR value.

Additionally, for example, the first device may determine power related to the PSFCH transmission based on the CBR value. Additionally, for example, the first device may perform the PSFCH transmission on the PSFCH resource based on the power related to the PSFCH transmission.

The proposed method can be applied to the device(s) described in various embodiments of the present disclosure. First, the processor 102 of the first device 100 may control the transceiver 106 to receive, from a second device, a physical sidelink control channel (PSCCH). In addition, the processor 102 of the first device 100 may control the transceiver 106 to receive, from the second device, a physical sidelink shared channel (PSSCH) related to the PSCCH. In addition, the processor 102 of the first device 100 may determine a physical sidelink feedback channel (PSFCH) resource related to the PSSCH based on a subchannel and a slot related to the PSSCH. In addition, the processor 102 of the first device 100 may obtain a channel busy ratio (CBR) value related to PSFCH transmission on the PSFCH resource. For example, the CBR value may be obtained based on measurement in a time interval between (i) a first slot before a first offset from a slot in which the PSSCH related to the PSFCH resource is received and (ii) a second slot before a second offset from the first slot.

Based on an embodiment of the present disclosure, a first device configured to perform wireless communication may be provided. For example, the first device may comprise: one or more memories storing instructions; one or more transceivers; and one or more processors connected to the one or more memories and the one or more transceivers. For example, the one or more processors may execute the instructions to: receive, from a second device, a physical sidelink control channel (PSCCH); receive, from the second device, a physical sidelink shared channel (PSSCH) related to the PSCCH; determine a physical sidelink feedback channel (PSFCH) resource related to the PSSCH based on a subchannel and a slot related to the PSSCH; and obtain a channel busy ratio (CBR) value related to PSFCH transmission on the PSFCH resource. For example, the CBR value may be obtained based on measurement in a time interval between (i) a first slot before a first offset from a slot in

which the PSSCH related to the PSFCH resource is received and (ii) a second slot before a second offset from the first slot.

Based on an embodiment of the present disclosure, an apparatus configured to control a first user equipment (UE) performing wireless communication may be provided. For example, the apparatus may comprise: one or more processors; and one or more memories operably connected to the one or more processors and storing instructions. For example, the one or more processors may execute the instructions to: receive, from a second UE, a physical sidelink control channel (PSCCH); receive, from the second UE, a physical sidelink shared channel (PSSCH) related to the PSCCH; determine a physical sidelink feedback channel (PSFCH) resource related to the PSSCH based on a sub-channel and a slot related to the PSSCH; and obtain a channel busy ratio (CBR) value related to PSFCH transmission on the PSFCH resource. For example, the CBR value may be obtained based on measurement in a time interval between (i) a first slot before a first offset from a slot in which the PSSCH related to the PSFCH resource is received and (ii) a second slot before a second offset from the first slot.

Based on an embodiment of the present disclosure, a non-transitory computer-readable storage medium storing instructions may be provided. For example, the instructions, when executed, may cause a first device to: receive, from a second device, a physical sidelink control channel (PSCCH); receive, from the second device, a physical sidelink shared channel (PSSCH) related to the PSCCH; determine a physical sidelink feedback channel (PSFCH) resource related to the PSSCH based on a subchannel and a slot related to the PSSCH; and obtain a channel busy ratio (CBR) value related to PSFCH transmission on the PSFCH resource. For example, the CBR value may be obtained based on measurement in a time interval between (i) a first slot before a first offset from a slot in which the PSSCH related to the PSFCH resource is received and (ii) a second slot before a second offset from the first slot.

Various embodiments of the present disclosure may be combined with each other.

Hereinafter, device(s) to which various embodiments of the present disclosure can be applied will be described.

The various descriptions, functions, procedures, proposals, methods, and/or operational flowcharts of the present disclosure described in this document may be applied to, without being limited to, a variety of fields requiring wireless communication/connection (e.g., 5G) between devices.

Hereinafter, a description will be given in more detail with reference to the drawings. In the following drawings/description, the same reference symbols may denote the same or corresponding hardware blocks, software blocks, or functional blocks unless described otherwise.

FIG. 19 shows a communication system 1, based on an embodiment of the present disclosure.

Referring to FIG. 19, a communication system 1 to which various embodiments of the present disclosure are applied includes wireless devices, Base Stations (BSs), and a network. Herein, the wireless devices represent devices performing communication using Radio Access Technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot 100a, vehicles 100b-1 and 100b-2, an extended Reality (XR) device 100c, a hand-held device 100d, a home appliance 100e, an Internet of Things (IoT) device 100f, and an Artificial Intelligence (AI) device/server



400. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The XR device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may be implemented as wireless devices and a specific wireless device 200a may operate as a B S/network node with respect to other wireless devices.

Here, wireless communication technology implemented in wireless devices 100a to 100f of the present disclosure may include Narrowband Internet of Things for low-power communication in addition to LTE, NR, and 6G. In this case, for example, NB-IoT technology may be an example of Low Power Wide Area Network (LPWAN) technology and may be implemented as standards such as LTE Cat NB1, and/or LTE Cat NB2, and is not limited to the name described above. Additionally or alternatively, the wireless communication technology implemented in the wireless devices 100a to 100f of the present disclosure may perform communication based on LTE-M technology. In this case, as an example, the LTE-M technology may be an example of the LPWAN and may be called by various names including enhanced Machine Type Communication (eMTC), and the like. For example, the LTE-M technology may be implemented as at least any one of various standards such as 1) LTE CAT 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-Bandwidth Limited (non-BL), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) LTE M, and is not limited to the name described above. Additionally or alternatively, the wireless communication technology implemented in the wireless devices 100a to 100f of the present disclosure may include at least one of Bluetooth, Low Power Wide Area Network (LPWAN), and ZigBee considering the low-power communication, and is not limited to the name described above. As an example, the ZigBee technology may generate personal area networks (PAN) related to small/low-power digital communication based on various standards including IEEE 802.15.4, and the like, and may be called by various names.

The wireless devices 100a to 100f may be connected to the network 300 via the BSs 200. An AI technology may be applied to the wireless devices 100a to 100f and the wireless devices 100a to 100f may be connected to the AI server 400 via the network 300. The network 300 may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices 100a to 100f may communicate with each other through the BSs 200/network 300, the wireless devices 100a to 100f may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles 100b-1 and 100b-2 may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). The IoT device

(e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices 100a to 100f.

Wireless communication/connections 150a, 150b, or 150c may be established between the wireless devices 100a to 100f/BS 200, or BS 200/BS 200. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication 150a, sidelink communication 150b (or, D2D communication), or inter BS communication (e.g., relay, Integrated Access Backhaul (IAB)). The wireless devices and the BSs/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections 150a and 150b. For example, the wireless communication/connections 150a and 150b may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

FIG. 20 shows wireless devices, based on an embodiment of the present disclosure.

Referring to FIG. 20, a first wireless device 100 and a second wireless device 200 may transmit radio signals through a variety of RATs (e.g., LTE and NR). Herein, {the first wireless device 100 and the second wireless device 200} may correspond to {the wireless device 100x and the BS 200} and/or {the wireless device 100x and the wireless device 100x} of FIG. 19.

The first wireless device 100 may include one or more processors 102 and one or more memories 104 and additionally further include one or more transceivers 106 and/or one or more antennas 108. The processor(s) 102 may control the memory(s) 104 and/or the transceiver(s) 106 and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) 102 may process information within the memory(s) 104 to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) 106. The processor(s) 102 may receive radio signals including second information/signals through the transceiver 106 and then store information obtained by processing the second information/signals in the memory(s) 104. The memory(s) 104 may be connected to the processor(s) 102 and may store a variety of information related to operations of the processor(s) 102. For example, the memory(s) 104 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 102 or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) 102 and the memory(s) 104 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 106 may be connected to the processor(s) 102 and transmit and/or receive radio signals through one or more antennas 108. Each of the transceiver(s) 106 may include a transmitter and/or a receiver. The transceiver(s) 106 may be interchangeably used with Radio Frequency (RF) unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

The second wireless device 200 may include one or more processors 202 and one or more memories 204 and addi-



tionally further include one or more transceivers **206** and/or one or more antennas **208**. The processor(s) **202** may control the memory(s) **204** and/or the transceiver(s) **206** and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) **202** may process information within the memory(s) **204** to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) **206**. The processor(s) **202** may receive radio signals including fourth information/signals through the transceiver(s) **106** and then store information obtained by processing the fourth information/signals in the memory(s) **204**. The memory(s) **204** may be connected to the processor(s) **202** and may store a variety of information related to operations of the processor(s) **202**. For example, the memory(s) **204** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **202** or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) **202** and the memory(s) **204** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **206** may be connected to the processor(s) **202** and transmit and/or receive radio signals through one or more antennas **208**. Each of the transceiver(s) **206** may include a transmitter and/or a receiver. The transceiver(s) **206** may be interchangeably used with RF unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices **100** and **200** will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors **102** and **202**. For example, the one or more processors **102** and **202** may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors **102** and **202** may generate one or more Protocol Data Units (PDUs) and/or one or more Service Data Unit (SDUs) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors **102** and **202** may generate messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors **102** and **202** may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document and provide the generated signals to the one or more transceivers **106** and **206**. The one or more processors **102** and **202** may receive the signals (e.g., baseband signals) from the one or more transceivers **106** and **206** and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document.

The one or more processors **102** and **202** may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more Application Specific Integrated Circuits (ASICs), one or more Digital Signal Processors (DSPs), one or more Digital

Signal Processing Devices (DSPDs), one or more Programmable Logic Devices (PLDs), or one or more Field Programmable Gate Arrays (FPGAs) may be included in the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be included in the one or more processors **102** and **202** or stored in the one or more memories **104** and **204** so as to be driven by the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by Read-Only Memories (ROMs), Random Access Memories (RAMs), Electrically Erasable Programmable Read-Only Memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of this document, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, through the one or more antennas **108** and **208**. In this document, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert

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the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters.

FIG. **21** shows a signal process circuit for a transmission signal, based on an embodiment of the present disclosure.

Referring to FIG. **21**, a signal processing circuit **1000** may include scramblers **1010**, modulators **1020**, a layer mapper **1030**, a precoder **1040**, resource mappers **1050**, and signal generators **1060**. An operation/function of FIG. **21** may be performed, without being limited to, the processors **102** and **202** and/or the transceivers **106** and **206** of FIG. **20**. Hardware elements of FIG. **21** may be implemented by the processors **102** and **202** and/or the transceivers **106** and **206** of FIG. **20**. For example, blocks **1010** to **1060** may be implemented by the processors **102** and **202** of FIG. **20**. Alternatively, the blocks **1010** to **1050** may be implemented by the processors **102** and **202** of FIG. **20** and the block **1060** may be implemented by the transceivers **106** and **206** of FIG. **20**.

Codewords may be converted into radio signals via the signal processing circuit **1000** of FIG. **21**. Herein, the codewords are encoded bit sequences of information blocks. The information blocks may include transport blocks (e.g., a UL-SCH transport block, a DL-SCH transport block). The radio signals may be transmitted through various physical channels (e.g., a PUSCH and a PDSCH).

Specifically, the codewords may be converted into scrambled bit sequences by the scramblers **1010**. Scramble sequences used for scrambling may be generated based on an initialization value, and the initialization value may include ID information of a wireless device. The scrambled bit sequences may be modulated to modulation symbol sequences by the modulators **1020**. A modulation scheme may include pi/2-Binary Phase Shift Keying (pi/2-BPSK), m-Phase Shift Keying (m-PSK), and m-Quadrature Amplitude Modulation (m-QAM). Complex modulation symbol sequences may be mapped to one or more transport layers by the layer mapper **1030**. Modulation symbols of each transport layer may be mapped (precoded) to corresponding antenna port(s) by the precoder **1040**. Outputs  $z$  of the precoder **1040** may be obtained by multiplying outputs  $y$  of the layer mapper **1030** by an  $N \times M$  precoding matrix  $W$ . Herein,  $N$  is the number of antenna ports and  $M$  is the number of transport layers. The precoder **1040** may perform precoding after performing transform precoding (e.g., DFT) for complex modulation symbols. Alternatively, the precoder **1040** may perform precoding without performing transform precoding.

The resource mappers **1050** may map modulation symbols of each antenna port to time-frequency resources. The time-frequency resources may include a plurality of symbols (e.g., a CP-OFDM symbols and DFT-s-OFDM symbols) in the time domain and a plurality of subcarriers in the frequency domain. The signal generators **1060** may generate radio signals from the mapped modulation symbols and the generated radio signals may be transmitted to other devices through each antenna. For this purpose, the signal generators **1060** may include Inverse Fast Fourier Transform (IFFT) modules, Cyclic Prefix (CP) inserters, Digital-to-Analog Converters (DACs), and frequency up-converters.

Signal processing procedures for a signal received in the wireless device may be configured in a reverse manner of the signal processing procedures **1010** to **1060** of FIG. **21**. For example, the wireless devices (e.g., **100** and **200** of FIG. **20**) may receive radio signals from the exterior through the

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antenna ports/transceivers. The received radio signals may be converted into baseband signals through signal restorers. To this end, the signal restorers may include frequency downlink converters, Analog-to-Digital Converters (ADCs), CP remover, and Fast Fourier Transform (FFT) modules. Next, the baseband signals may be restored to codewords through a resource demapping procedure, a postcoding procedure, a demodulation processor, and a descrambling procedure. The codewords may be restored to original information blocks through decoding. Therefore, a signal processing circuit (not illustrated) for a reception signal may include signal restorers, resource demappers, a postcoder, demodulators, descramblers, and decoders.

FIG. **22** shows another example of a wireless device, based on an embodiment of the present disclosure. The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. **19**).

Referring to FIG. **22**, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. **20** and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** and/or the one or more memories **104** and **204** of FIG. **20**. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** and/or the one or more antennas **108** and **208** of FIG. **20**. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the wireless device based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit, a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100a** of FIG. **19**), the vehicles (**100b-1** and **100b-2** of FIG. **19**), the XR device (**100c** of FIG. **19**), the hand-held device (**100d** of FIG. **19**), the home appliance (**100e** of FIG. **19**), the IoT device (**100f** of FIG. **19**), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. **19**), the BSs (**200** of FIG. **19**), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

In FIG. **22**, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by

wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor, an Electronic Control Unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a Random Access Memory (RAM), a Dynamic RAM (DRAM), a Read Only Memory (ROM), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

Hereinafter, an example of implementing FIG. 22 will be described in detail with reference to the drawings.

FIG. 23 shows a hand-held device, based on an embodiment of the present disclosure. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), or a portable computer (e.g., a notebook). The hand-held device may be referred to as a mobile station (MS), a user terminal (UT), a Mobile Subscriber Station (MSS), a Subscriber Station (SS), an Advanced Mobile Station (AMS), or a Wireless Terminal (WT).

Referring to FIG. 23, a hand-held device **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a memory unit **130**, a power supply unit **140a**, an interface unit **140b**, and an I/O unit **140c**. The antenna unit **108** may be configured as a part of the communication unit **110**. Blocks **110** to **130/140a** to **140c** correspond to the blocks **110** to **130/140** of FIG. 22, respectively.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from other wireless devices or BSs. The control unit **120** may perform various operations by controlling constituent elements of the hand-held device **100**. The control unit **120** may include an Application Processor (AP). The memory unit **130** may store data/parameters/programs/code/commands needed to drive the hand-held device **100**. The memory unit **130** may store input/output data/information. The power supply unit **140a** may supply power to the hand-held device **100** and include a wired/wireless charging circuit, a battery, etc. The interface unit **140b** may support connection of the hand-held device **100** to other external devices. The interface unit **140b** may include various ports (e.g., an audio I/O port and a video I/O port) for connection with external devices. The I/O unit **140c** may input or output video information/signals, audio information/signals, data, and/or information input by a user. The I/O unit **140c** may include a camera, a microphone, a user input unit, a display unit **140d**, a speaker, and/or a haptic module.

As an example, in the case of data communication, the I/O unit **140c** may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user and the acquired information/signals may be stored in the memory unit **130**. The communication unit **110** may convert the information/signals stored in the memory into radio signals and transmit the converted radio signals to other wireless devices directly or to a BS. The communication unit **110** may receive radio signals from other wireless devices or the BS and then restore the received radio signals into original information/signals. The restored information/signals may be stored in the memory unit **130** and may be output as various types (e.g., text, voice, images, video, or haptic) through the I/O unit **140c**.

FIG. 24 shows a vehicle or an autonomous vehicle, based on an embodiment of the present disclosure. The vehicle or autonomous vehicle may be implemented by a mobile robot, a car, a train, a manned/unmanned Aerial Vehicle (AV), a ship, etc.

Referring to FIG. 24, a vehicle or autonomous vehicle **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a driving unit **140a**, a power supply unit **140b**, a sensor unit **140c**, and an autonomous driving unit **140d**. The antenna unit **108** may be configured as a part of the communication unit **110**. The blocks **110/130/140a** to **140d** correspond to the blocks **110/130/140** of FIG. 22, respectively.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles, BSs (e.g., gNBs and road side units), and servers. The control unit **120** may perform various operations by controlling elements of the vehicle or the autonomous vehicle **100**. The control unit **120** may include an Electronic Control Unit (ECU). The driving unit **140a** may cause the vehicle or the autonomous vehicle **100** to drive on a road. The driving unit **140a** may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The power supply unit **140b** may supply power to the vehicle or the autonomous vehicle **100** and include a wired/wireless charging circuit, a battery, etc. The sensor unit **140c** may acquire a vehicle state, ambient environment information, user information, etc. The sensor unit **140c** may include an Inertial Measurement Unit (IMU) sensor, a collision sensor, a wheel sensor, a speed sensor, a slope sensor, a weight sensor, a heading sensor, a position module, a vehicle forward/backward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illumination sensor, a pedal position sensor, etc. The autonomous driving unit **140d** may implement technology for maintaining a lane on which a vehicle is driving, technology for automatically adjusting speed, such as adaptive cruise control, technology for autonomously driving along a determined path, technology for driving by automatically setting a path if a destination is set, and the like.

For example, the communication unit **110** may receive map data, traffic information data, etc. from an external server. The autonomous driving unit **140d** may generate an autonomous driving path and a driving plan from the obtained data. The control unit **120** may control the driving unit **140a** such that the vehicle or the autonomous vehicle **100** may move along the autonomous driving path according to the driving plan (e.g., speed/direction control). In the middle of autonomous driving, the communication unit **110** may aperiodically/periodically acquire recent traffic information data from the external server and acquire surrounding traffic information data from neighboring vehicles. In the middle of autonomous driving, the sensor unit **140c** may obtain a vehicle state and/or surrounding environment information. The autonomous driving unit **140d** may update the autonomous driving path and the driving plan based on the newly obtained data/information. The communication unit **110** may transfer information about a vehicle position, the autonomous driving path, and/or the driving plan to the external server. The external server may predict traffic information data using AI technology, etc., based on the information collected from vehicles or autonomous vehicles and provide the predicted traffic information data to the vehicles or the autonomous vehicles.

Claims in the present description can be combined in a various way. For instance, technical features in method

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claims of the present description can be combined to be implemented or performed in an apparatus, and technical features in apparatus claims can be combined to be implemented or performed in a method. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in an apparatus. Further, technical features in method claim(s) and apparatus claim(s) can be combined to be implemented or performed in a method.

What is claimed is:

1. A method comprising:

receiving, by a first device, from a second device, a physical sidelink control channel;

receiving, by the first device, from the second device, a physical sidelink shared channel related to the physical sidelink control channel;

determining, by the first device, a physical sidelink feedback channel resource related to the physical sidelink shared channel based on a subchannel and a slot related to the physical sidelink shared channel; and

obtaining, by the first device, a channel busy ratio value related to a physical sidelink feedback channel transmission on the physical sidelink feedback channel resource,

wherein the channel busy ratio value related to the physical sidelink feedback channel transmission is obtained based on a measurement for physical sidelink control channel resource domain, physical sidelink shared channel resource domain, and physical sidelink feedback channel resource domain in a time interval between:

(i) a first slot before a first offset from a slot in which the physical sidelink shared channel related to the physical sidelink feedback channel resource is received, and

(ii) a second slot before a second offset from the first slot.

2. The method of claim 1, wherein, based on a plurality of physical sidelink shared channels related to the physical sidelink feedback channel resource being received in a plurality of slots, the slot in which the physical sidelink shared channel is received is a last slot among the plurality of slots.

3. The method of claim 1, wherein, based on a plurality of physical sidelink shared channels related to the physical sidelink feedback channel resource being received in a plurality of slots, the slot in which the physical sidelink shared channel is received is a 1st slot among the plurality of slots.

4. The method of claim 1, wherein, based on a plurality of physical sidelink shared channels related to the physical sidelink feedback channel resource being received in a plurality of slots, the slot in which the physical sidelink shared channel is received is a slot in a pre-configured order for the first device among the plurality of slots.

5. The method of claim 1, wherein, based on (i) a plurality of physical sidelink control channels related to the physical sidelink feedback channel resource being transmitted in a plurality of slots, and (ii) successful decoding of at least one physical sidelink control channel in at least one slot among the plurality of slots by the first device, the slot in which the physical sidelink shared channel is received is a last slot among the at least one slot related to the at least one physical sidelink control channel.

6. The method of claim 1, wherein, based on (i) a plurality of physical sidelink control channels related to the physical sidelink feedback channel resource being transmitted in a

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plurality of slots, and (ii) successful decoding of at least one physical sidelink control channel in at least one slot among the plurality of slots by the first device, the slot in which the physical sidelink shared channel is received is a 1st slot among the at least one slot related to the at least one physical sidelink control channel.

7. The method of claim 1, wherein the first offset is an offset related to a time required for processing of the channel busy ratio value or an offset related to a time required for power setting for the physical sidelink feedback channel transmission, and

wherein the second offset is an offset for determining a length of a time window related to a channel busy ratio measurement.

8. The method of claim 1, wherein the first offset or the second offset for obtaining the channel busy ratio value related to the physical sidelink feedback channel transmission and an offset for obtaining a channel busy ratio value related to a physical sidelink control channel transmission or a physical sidelink shared channel transmission are independently configured for the first device, respectively.

9. The method of claim 1, wherein the first offset or the second offset is configured differently based on at least one of a numerology, a frequency range, a service type, a service priority, a quality of service requirement, a cast type, or a traffic type.

10. The method of claim 1, further comprising: determining a parameter related to the physical sidelink feedback channel transmission based on the channel busy ratio value.

11. The method of claim 1, further comprising: determining power related to the physical sidelink feedback channel transmission based on the channel busy ratio value; and

performing the physical sidelink feedback channel transmission on the physical sidelink feedback channel resource based on the power related to the physical sidelink feedback channel transmission.

12. A first device comprising:

at least one transceiver;

at least one processor; and

at least one memory connected to the at least one processor and storing instruction that, upon being executed, cause the first device to perform operation comprising:

receiving, from a second device, a physical sidelink control channel;

receiving, from the second device, a physical sidelink shared channel related to the physical sidelink control channel;

determining a physical sidelink feedback channel resource related to the physical sidelink shared channel based on a subchannel and a slot related to the physical sidelink shared channel; and

obtaining a channel busy ratio value related to a physical sidelink feedback channel transmission on the physical sidelink feedback channel resource,

wherein the channel busy ratio value related to the physical sidelink feedback channel transmission is obtained based on a measurement for physical sidelink control channel resource domain, physical sidelink shared channel resource domain, and physical sidelink feedback channel resource domain in a time interval between:

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- (i) a first slot before a first offset from a slot in which the physical sidelink shared channel related to the physical sidelink feedback channel resource is received, and

- (ii) a second slot before a second offset from the first slot.

13. The first device of claim 12, wherein, based on a plurality of physical sidelink shared channels related to the physical sidelink feedback channel resource being received in a plurality of slots, the slot in which the physical sidelink shared channel is received is a last slot among the plurality of slots.

14. The first device of claim 12, wherein, based on a plurality of physical sidelink shared channels related to the physical sidelink feedback channel resource being received in a plurality of slots, the slot in which the physical sidelink shared channel is received is a 1st slot among the plurality of slots.

15. An apparatus configured to control a first user equipment performing wireless communication, the apparatus comprising:

at least one processor; and

at least one memory connected to the at least one processor and storing instruction that, upon being executed, cause the first user equipment to perform operation comprising:

receiving, from a second user equipment, a physical sidelink control channel;

receiving, from the second user equipment, a physical sidelink shared channel related to the physical sidelink control channel;

determining a physical sidelink feedback channel resource related to the physical sidelink shared channel

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based on a subchannel and a slot related to the physical sidelink shared channel; and

obtaining a channel busy ratio value related to a physical sidelink feedback channel transmission on the physical sidelink feedback channel resource,

wherein the channel busy ratio value related to the physical sidelink feedback channel transmission is obtained based on a measurement for physical sidelink control channel resource domain, physical sidelink shared channel resource domain, and physical sidelink feedback channel resource domain in a time interval between:

- (i) a first slot before a first offset from a slot in which the physical sidelink shared channel related to the physical sidelink feedback channel resource is received, and

- (ii) a second slot before a second offset from the first slot.

16. The apparatus of claim 15, wherein, based on a plurality of physical sidelink shared channels related to the physical sidelink feedback channel resource being received in a plurality of slots, the slot in which the physical sidelink shared channel is received is a last slot among the plurality of slots.

17. The apparatus of claim 15, wherein, based on a plurality of physical sidelink shared channels related to the physical sidelink feedback channel resource being received in a plurality of slots, the slot in which the physical sidelink shared channel is received is a 1st slot among the plurality of slots.

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